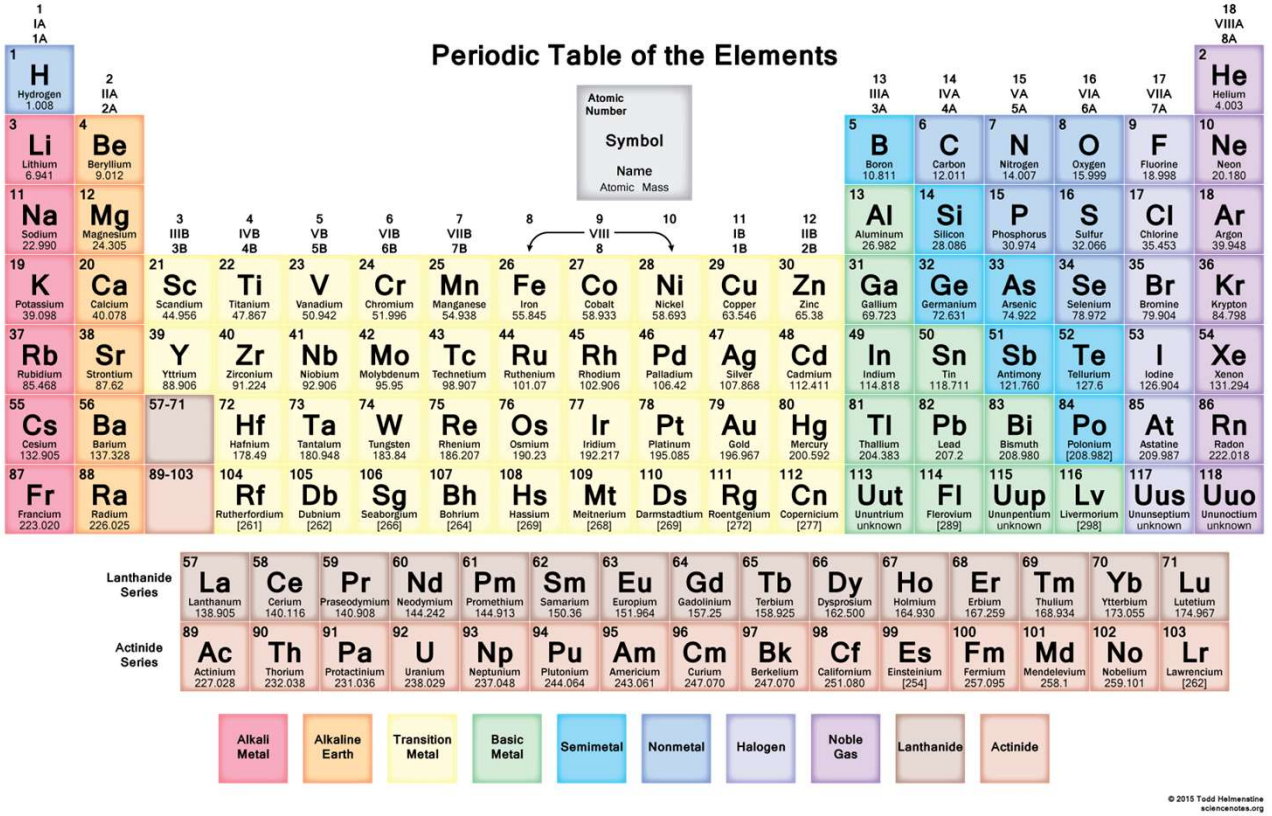
Chemistry of the Earth, Lecture 2:

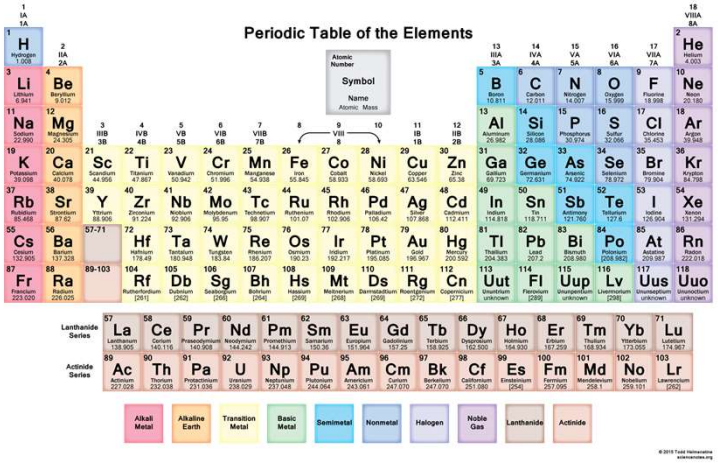
The Periodic Table – A Geochemist's View

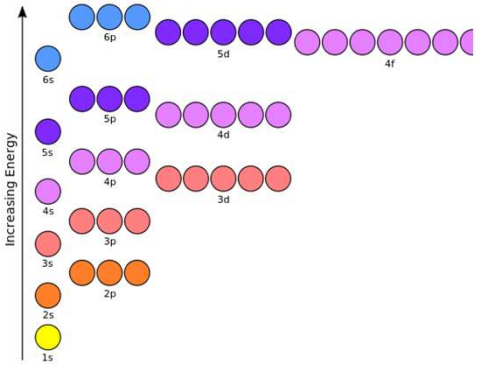


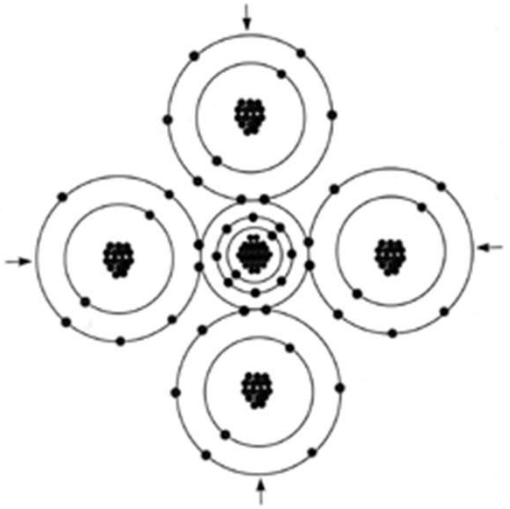
Part 1: Groups, Periods and Electrons

OBJECTIVES OF THIS WEEK'S LECTURE and PRACTICAL

- Review the Periodic Table of the Elements – Groups and Periods
- Understand electron configurations and why they are important
- Understand why groups of elements behave in similar ways
- Review the basic types of bonding important to minerals
- Introduction to the Geochemist's Periodic Table







THIS WEEK'S READING

Basic:

Grotzinger & Jordan **(Basic: Minimum reading)**	**Bonding and ions: 55-58** **Appendix 3: Chemical reactions**
Marshak	Chapter 5 (especially Box 5.1)

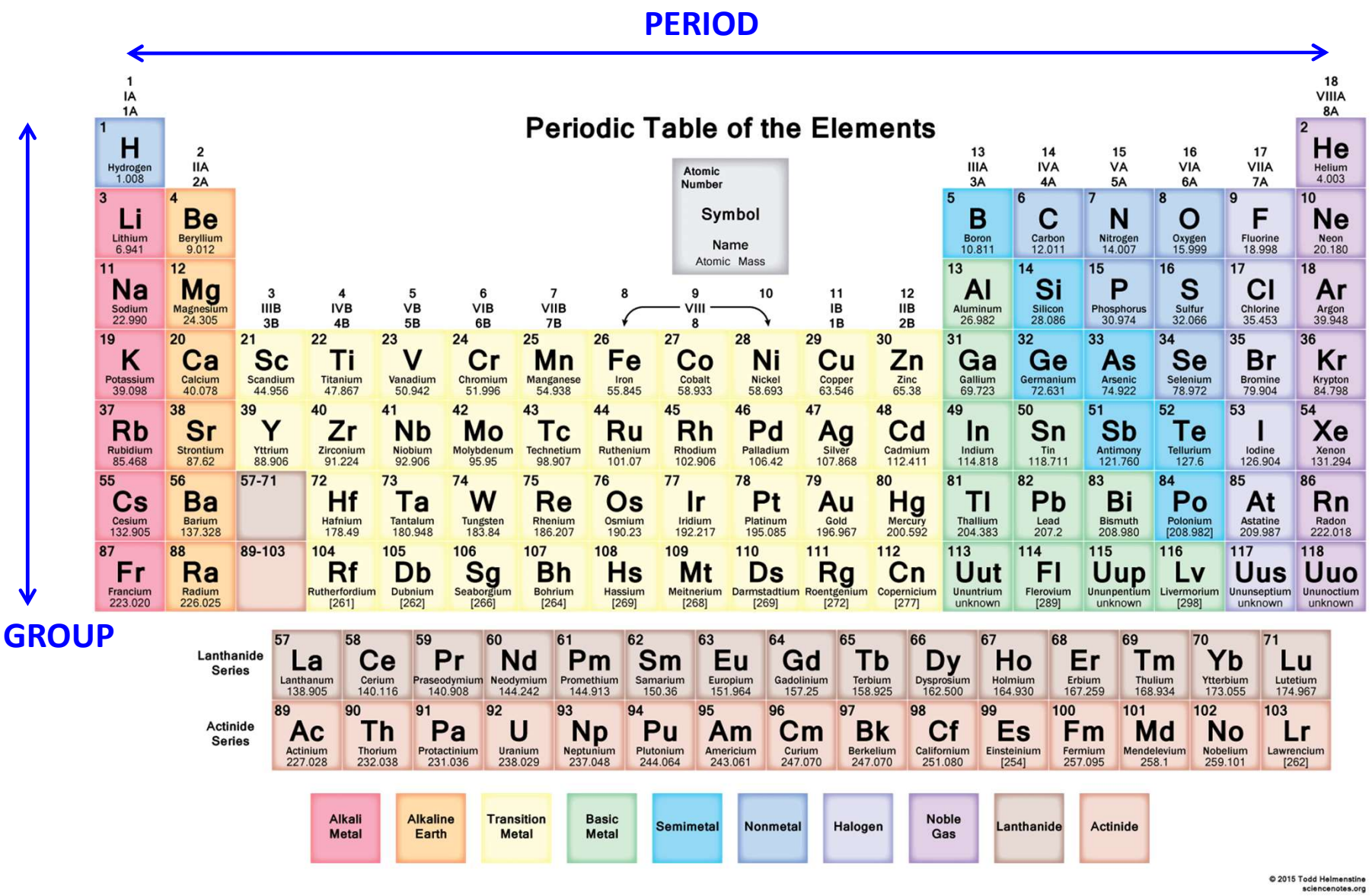
More detailed:

McSween & Richardson	**Chapter 2, How Elements Behave**

More advanced:

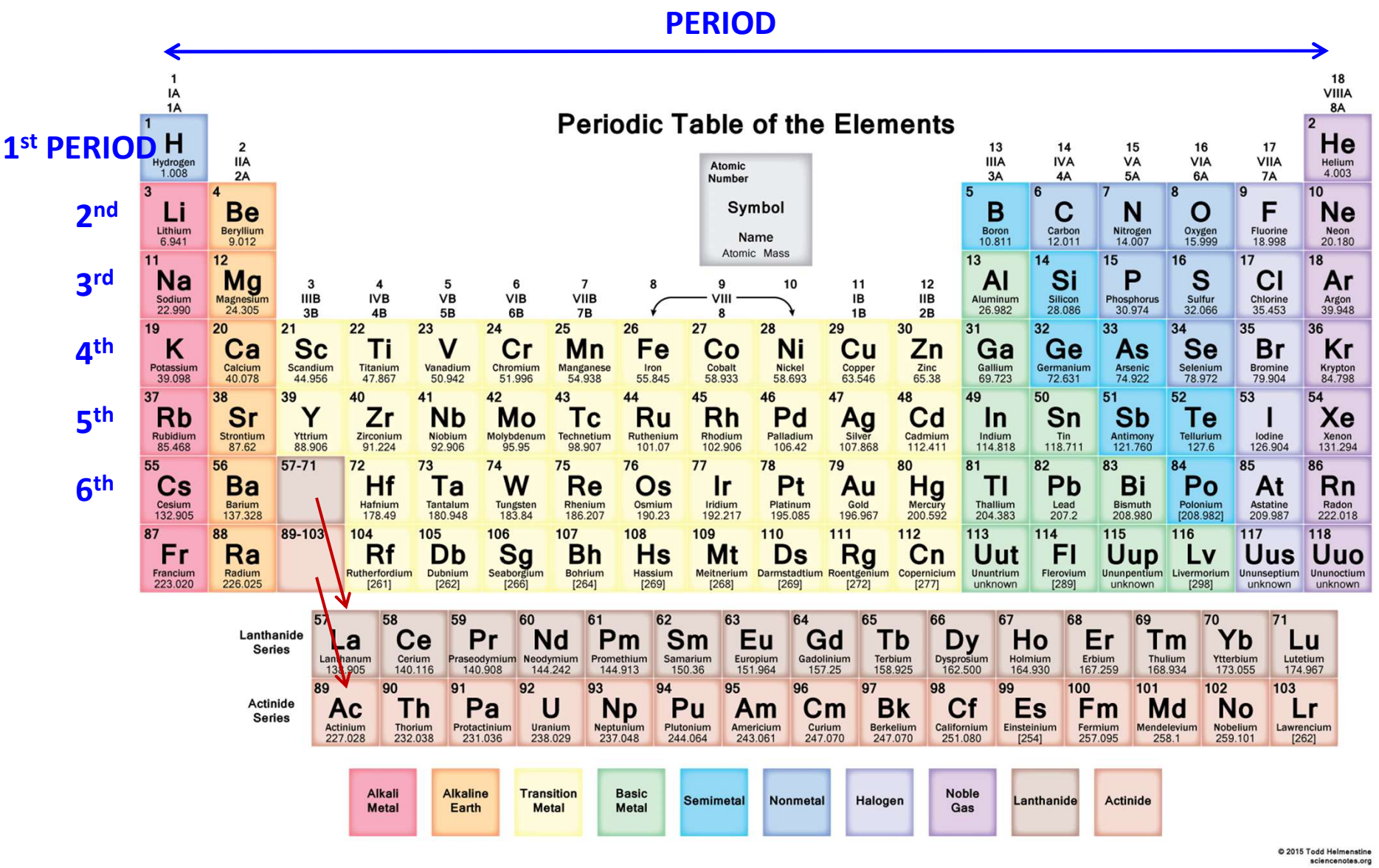
Gill	Chapters 5, 6 and 7
Misra	Chapters 2 and 3
White	Chapters 1.4 and 7 (Figures 1.3 and 1.5 have errors – see slides)

PERIODIC TABLE: GROUPS AND PERIODS



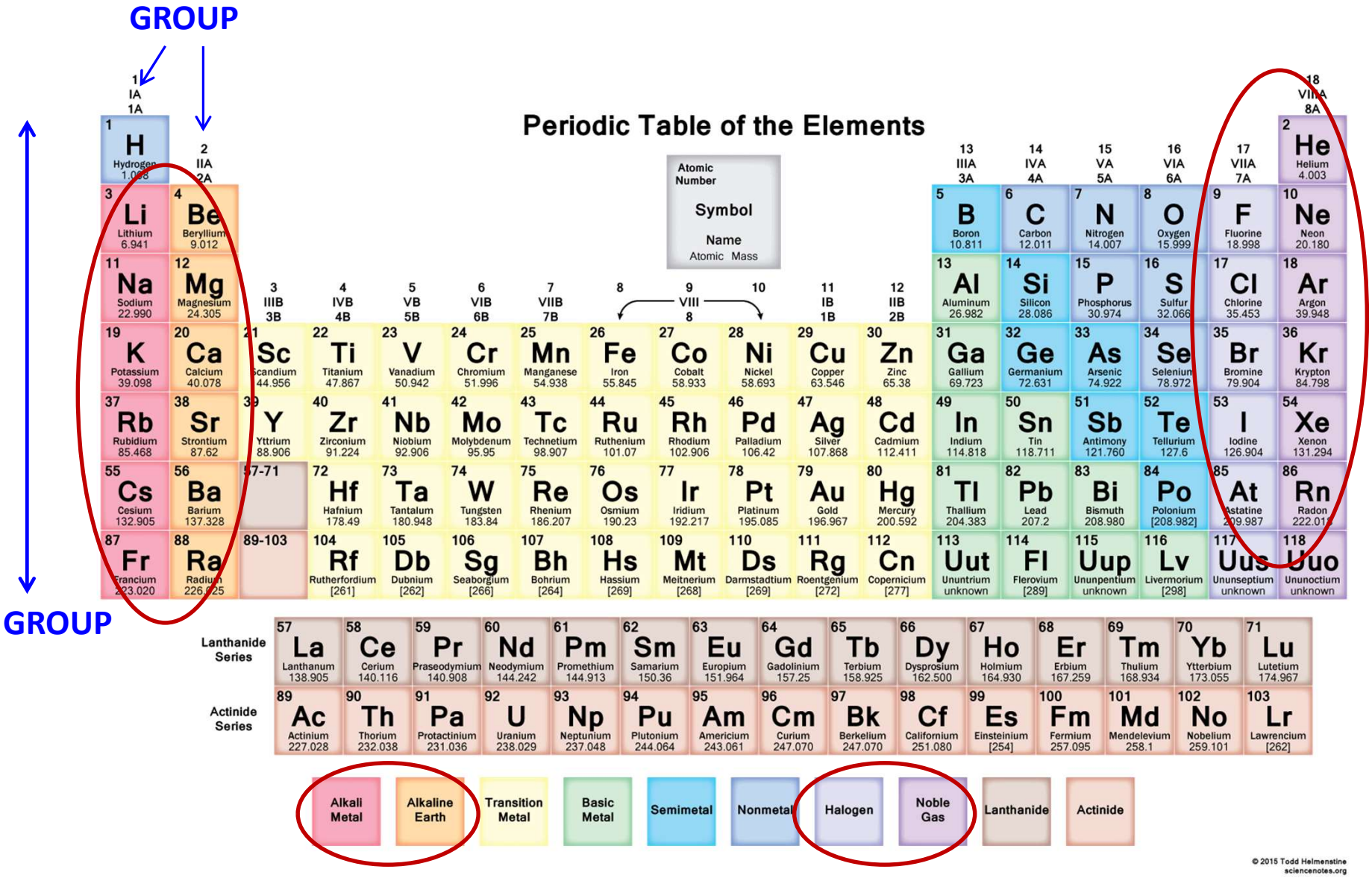
http://sciencenotes.org/printable-periodic-table/

The Table is a list of elements with increasing Atomic Number (Z), across periods



http://sciencenotes.org/printable-periodic-table/

Groups are vertical columns - elements in groups are chemically similar.
Groups include: Alkali metal, Alkaline Earth, Halogens, Noble Gases



http://sciencenotes.org/printable-periodic-table/

ELECTRON CONFIGURATION

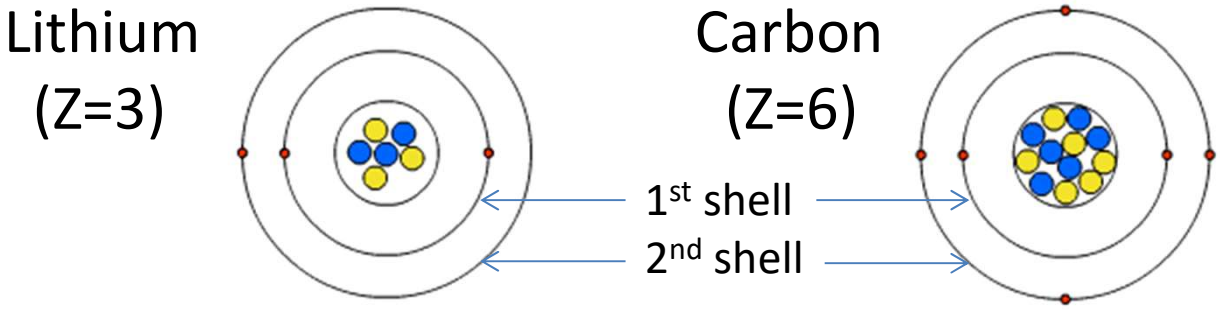
The chemical properties of an element depend on the way its electrons are arranged around the nucleus – electron configuration

For a neutral atom, Number of electrons = Number of protons (= Z)

- Electrons are arranged in **SHELLS** around the nucleus
- Each shell has one or more **SUBSHELLS**
- Each subshell is composed of one or more **ORBITALS**
- Each orbital contains a maximum of two electrons

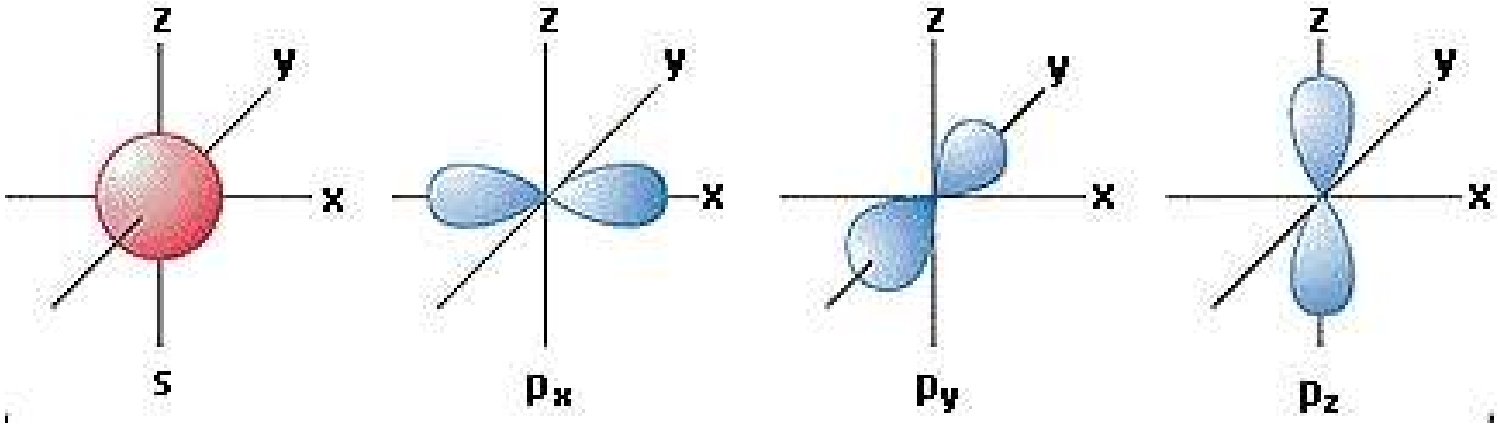
SHELL	SUBSHELL	Number of ORBITALS in subshell	Maximum number of ELECTRONS	Total electrons in shell
1 (K)	1s	1	2	2
2 (L)	2s	1	2	8
	2p	3	6	
3 (M)	3s	1	2	18
	3p	3	6	
	3d	5	10	
4 (N)	4s	1	2	32
	4p	3	6	
	4d	5	10	
	4f	7	14	

Simple view of atom – electrons are in layered shells



http://www.bbc.co.uk/schools/gcsebitesize

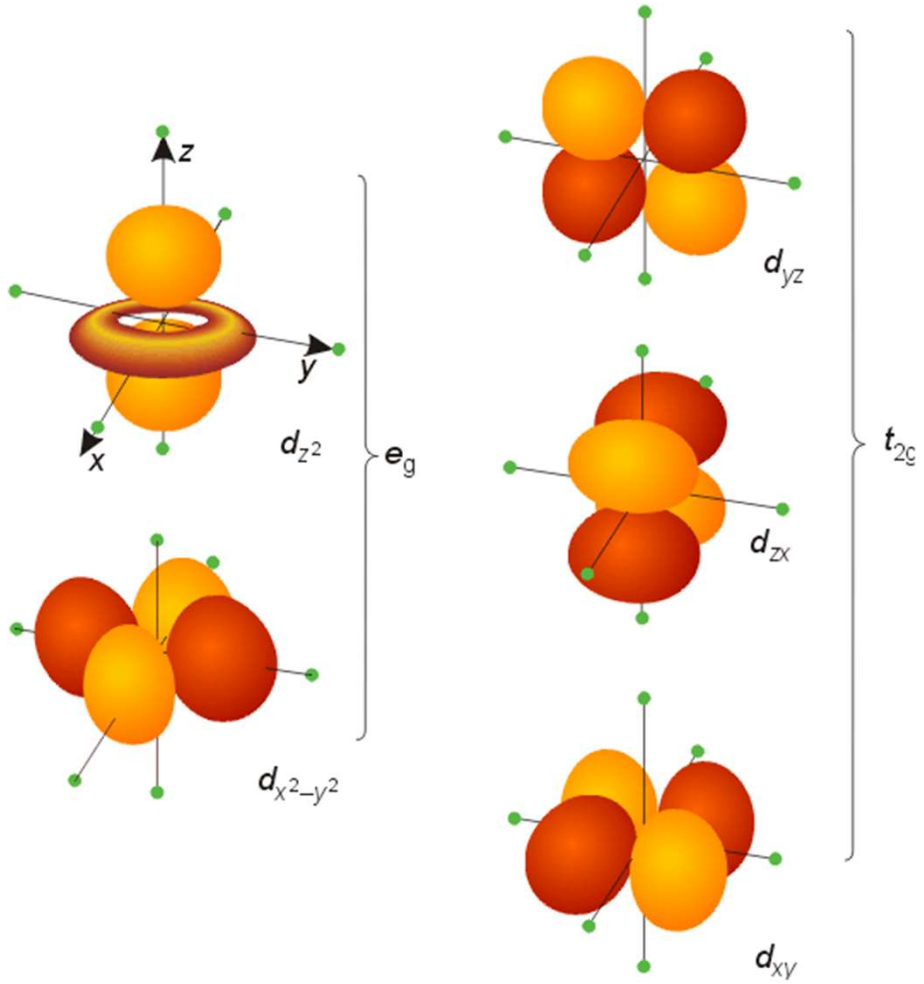
More detailed view – electrons are in orbitals



- **s orbitals** are spherical

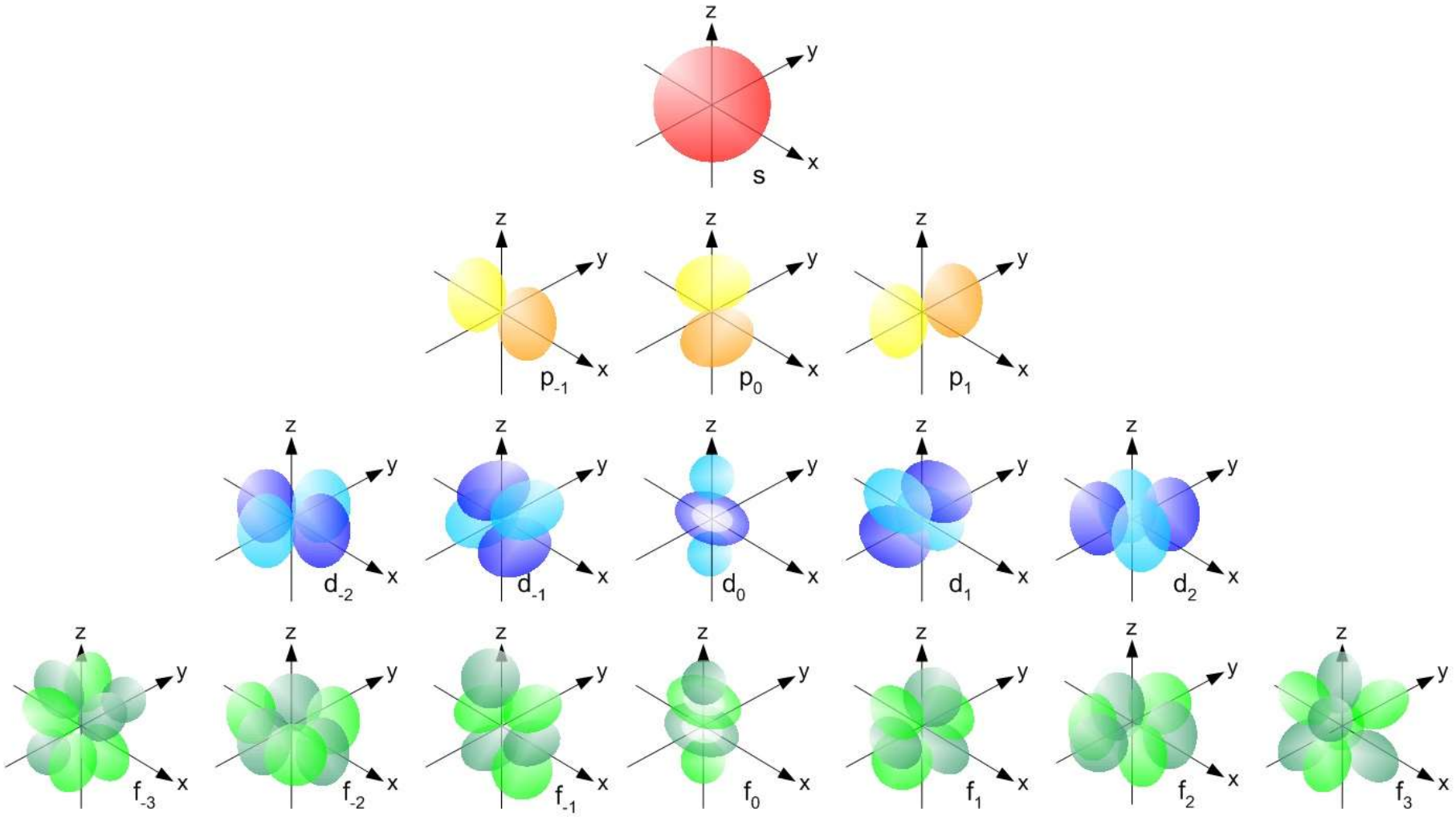
- 3 possible **p orbitals** in each p subshell
- each p orbital consist of two lobes
- each p orbital is aligned with a different axis (x, y, z)

http://perso.numericable.fr/vincent.hedberg/atomic_theory/2_atomic_theory_hl.htm



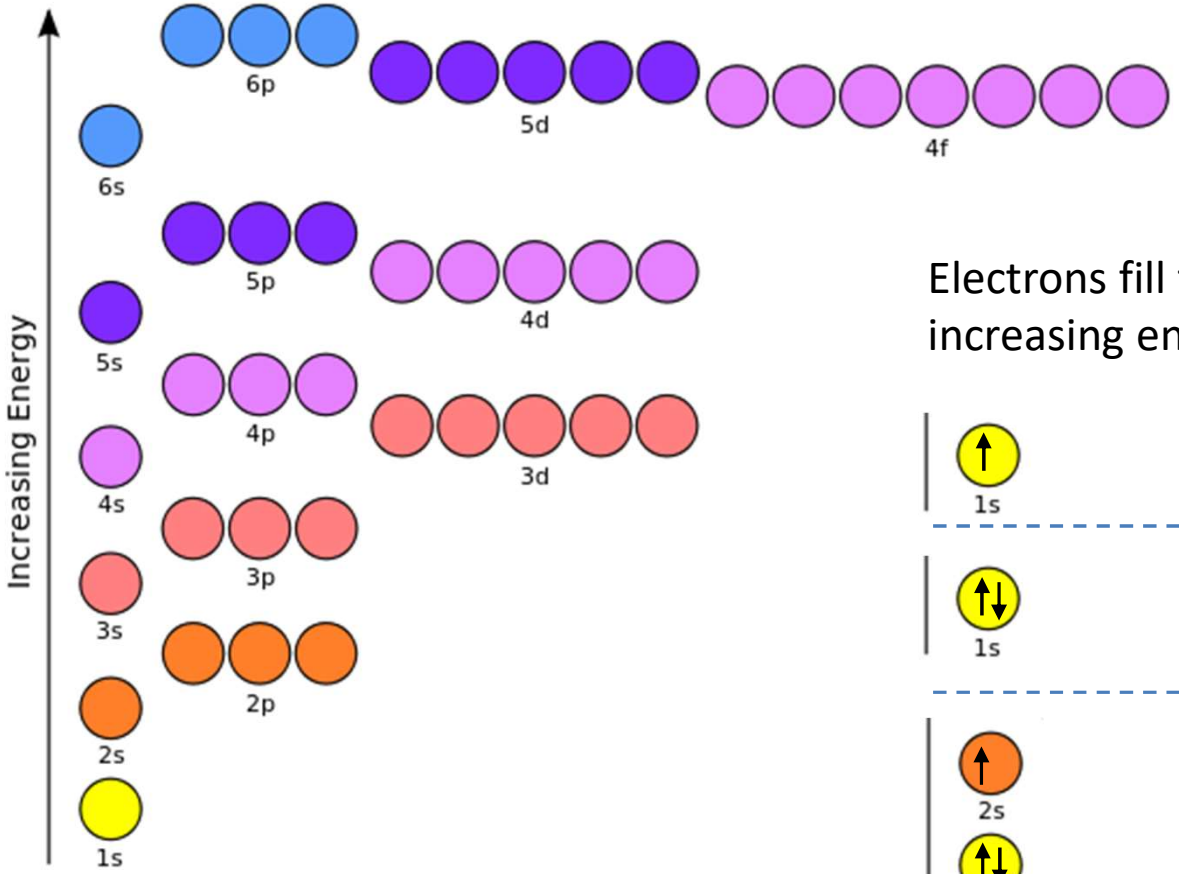
- five possible **d orbitals** in each d subshell
- each d orbital is aligned in a different direction relative to x, y, z axes
- four sets of d orbitals consist of four lobes
- the fifth d orbital (dz^2) has a unique shape

http://perso.numericable.fr/vincent.hedberg/atomic_theory/2_atomic_theory_hl.htm

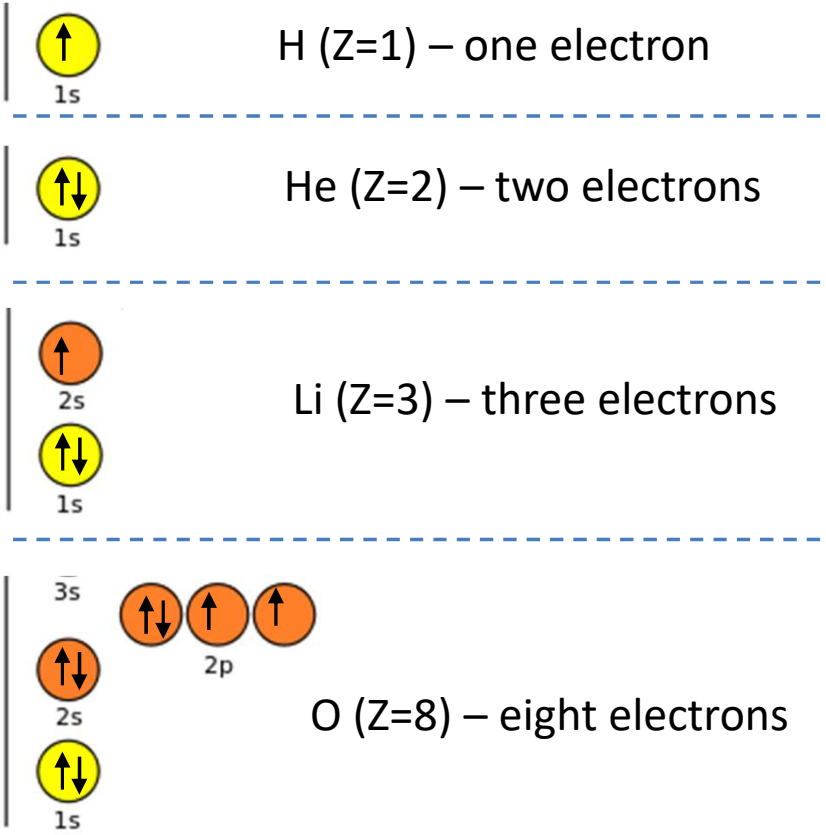


- seven possible **f orbitals** in each f subshell
- shapes are complicated
- don’t worry about the orbital names in this figure

Electron orbitals have different energy levels:

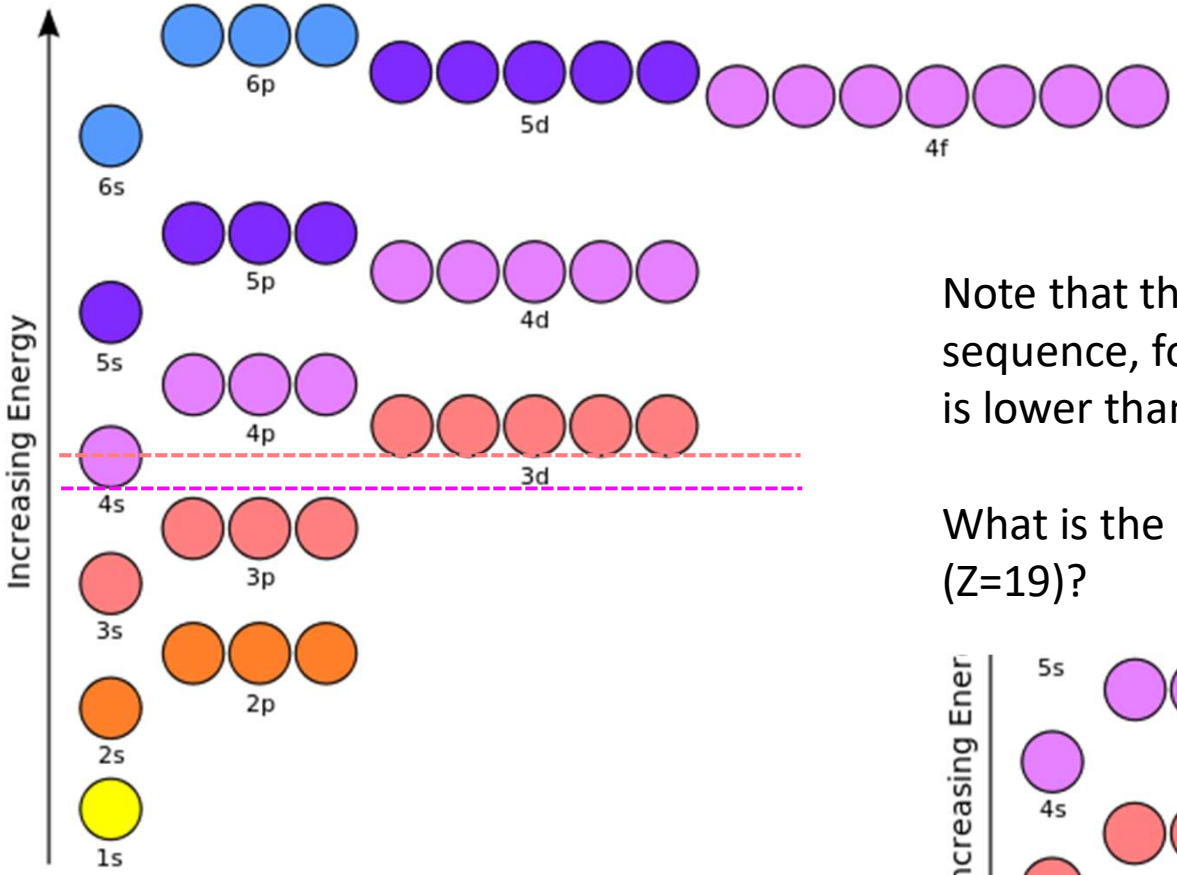


Electrons fill the orbitals in order of increasing energy:



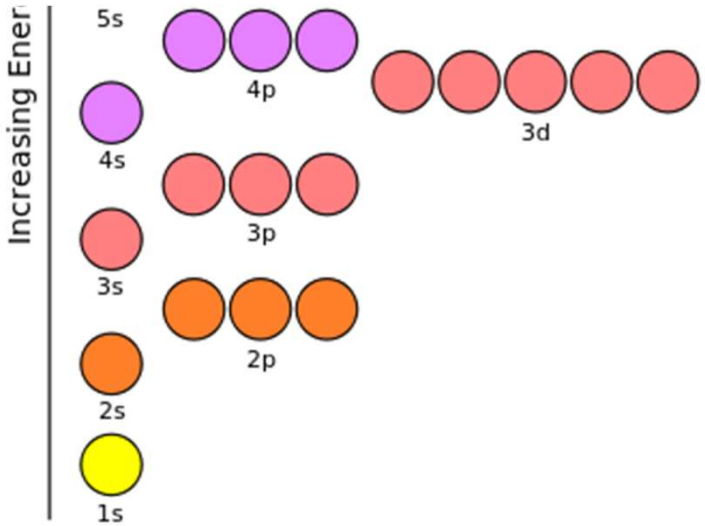
https://en.wikibooks.org/wiki/High_School_Chemistry/Electron_Configurations

Electron orbitals have different energy levels:



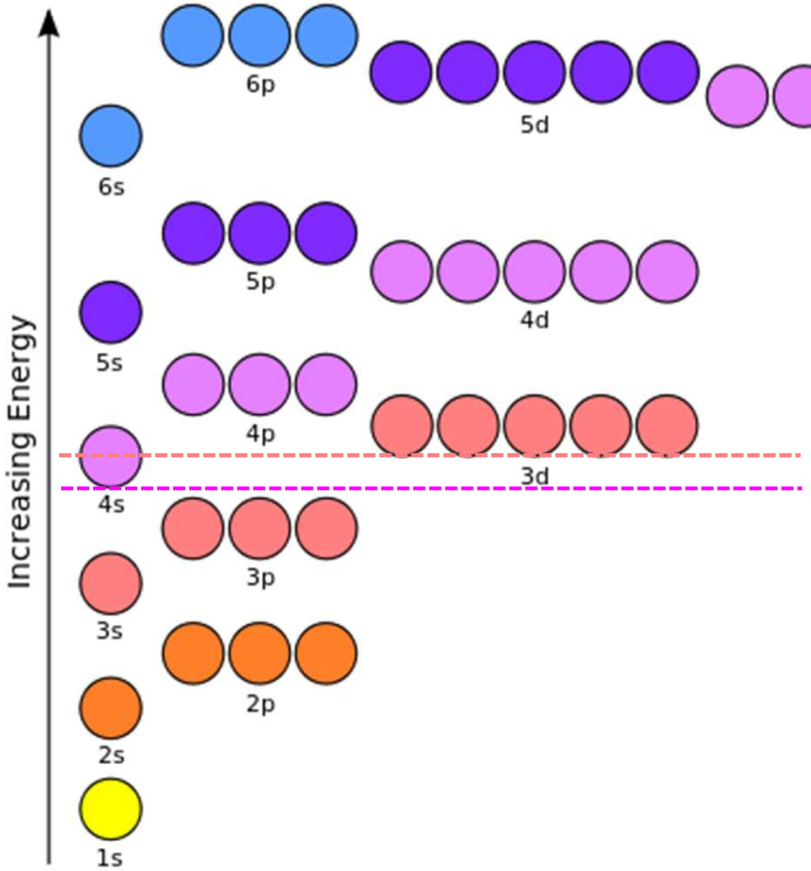
Note that the shells are not filled in sequence, for example the energy of 4s is lower than the energy of 3d.

What is the electron configuration for K (Z=19)?



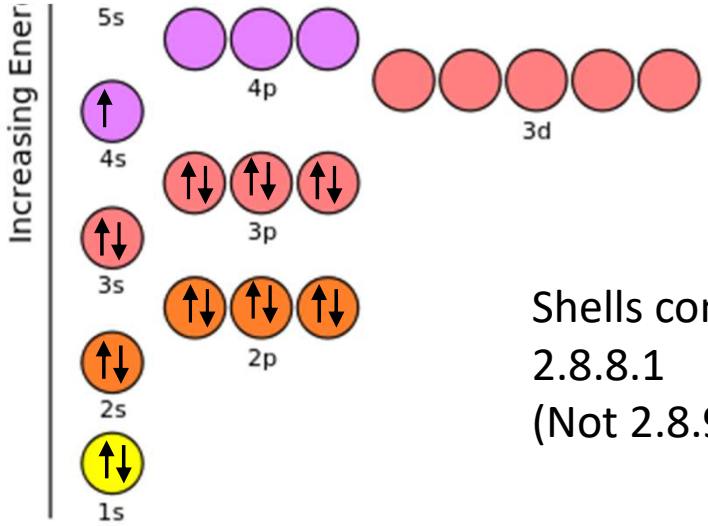
https://en.wikibooks.org/wiki/High_School_Chemistry/Electron_Configurations

Electron orbitals have different energy levels:



Note that the shells are not filled in sequence, for example the energy of 4s is lower than the energy of 3d.

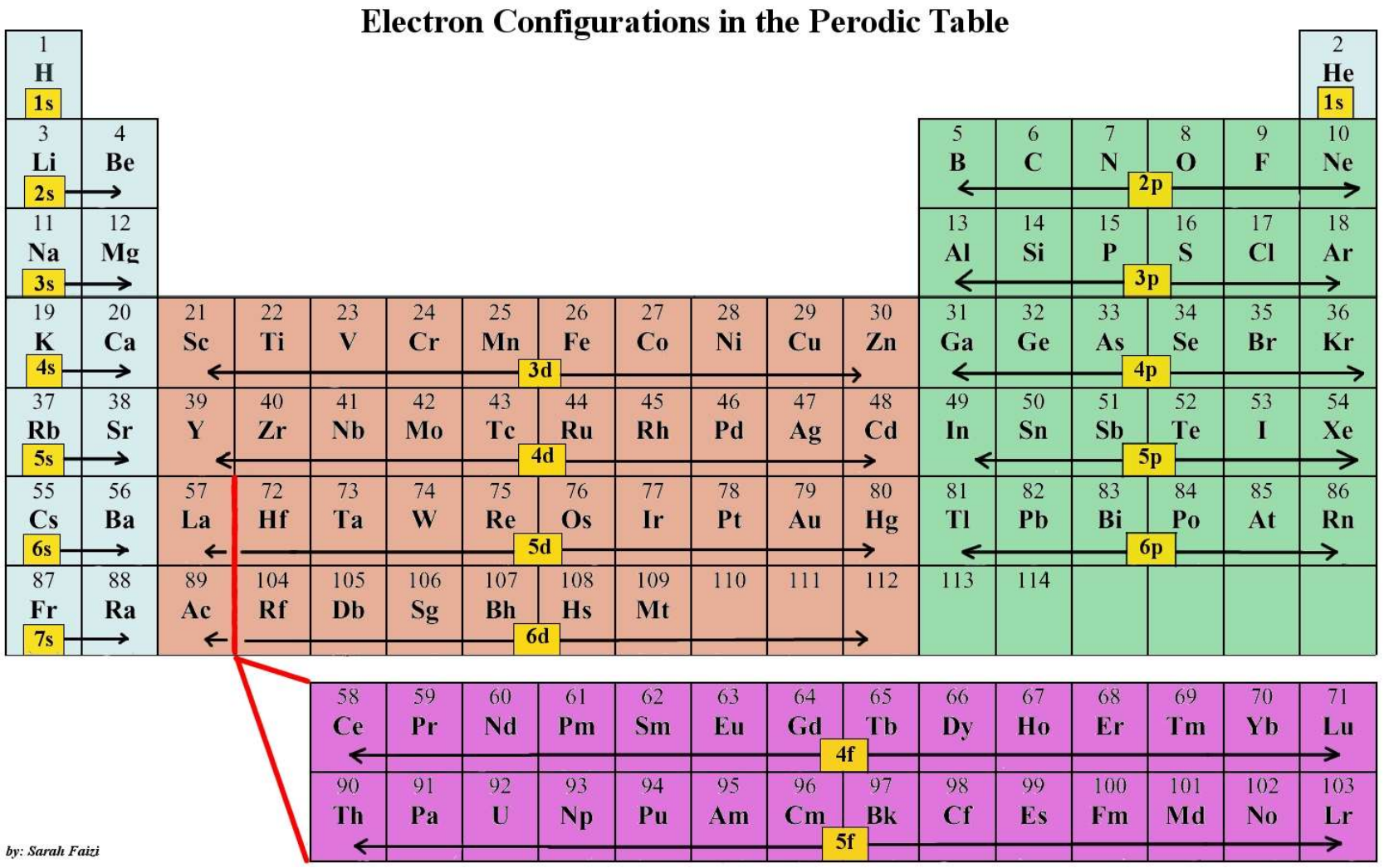
Configuration for K (Z=19):



Shells contain
2.8.8.1
(Not 2.8.9)

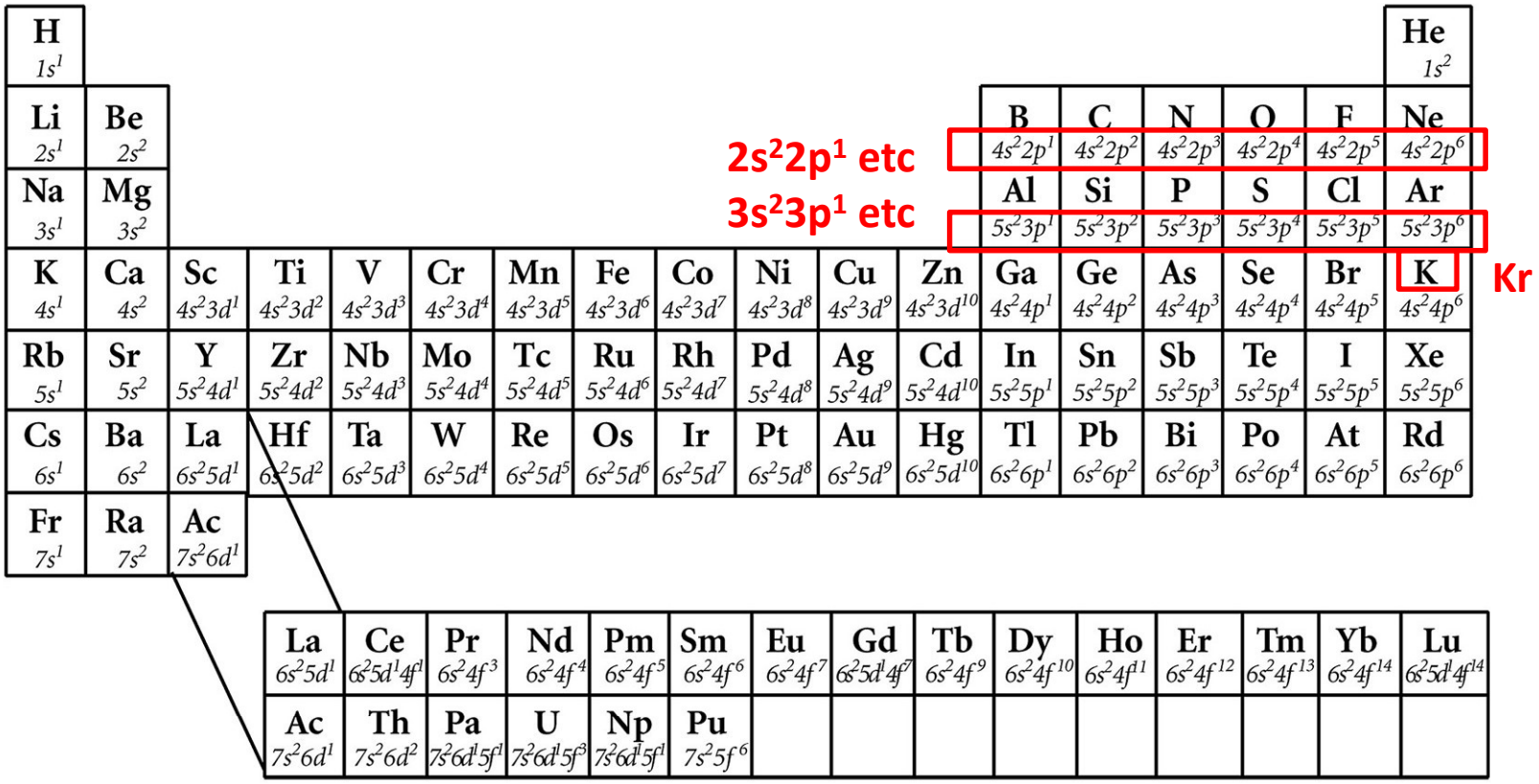
https://en.wikibooks.org/wiki/High_School_Chemistry/Electron_Configurations

The arrangement of elements in the Periodic Table follows the electron configurations. Yellow boxes e.g. 2p below refer to the orbitals that are being filled. Orbitals with lower energy levels are assumed to be full.



http://chemwiki.ucdavis.edu/Core/Inorganic_Chemistry/Electronic_Structure_of_Atoms_and_Molecules/Electronic_Configurations

Geochemistry by White - **errors**



H $1s^1$																	He $1s^2$
Li $2s^1$	Be $2s^2$											B $4s^2 2p^1$	C $4s^2 2p^2$	N $4s^2 2p^3$	O $4s^2 2p^4$	F $4s^2 2p^5$	Ne $4s^2 2p^6$
Na $3s^1$	Mg $3s^2$											Al $5s^2 3p^1$	Si $5s^2 3p^2$	P $5s^2 3p^3$	S $5s^2 3p^4$	Cl $5s^2 3p^5$	Ar $5s^2 3p^6$
K $4s^1$	Ca $4s^2$	Sc $4s^2 3d^1$	Ti $4s^2 3d^2$	V $4s^2 3d^3$	Cr $4s^2 3d^4$	Mn $4s^2 3d^5$	Fe $4s^2 3d^6$	Co $4s^2 3d^7$	Ni $4s^2 3d^8$	Cu $4s^2 3d^9$	Zn $4s^2 3d^{10}$	Ga $4s^2 4p^1$	Ge $4s^2 4p^2$	As $4s^2 4p^3$	Se $4s^2 4p^4$	Br $4s^2 4p^5$	K $4s^2 4p^6$
Rb $5s^1$	Sr $5s^2$	Y $5s^2 4d^1$	Zr $5s^2 4d^2$	Nb $5s^2 4d^3$	Mo $5s^2 4d^4$	Tc $5s^2 4d^5$	Ru $5s^2 4d^6$	Rh $5s^2 4d^7$	Pd $5s^2 4d^8$	Ag $5s^2 4d^9$	Cd $5s^2 4d^{10}$	In $5s^2 5p^1$	Sn $5s^2 5p^2$	Sb $5s^2 5p^3$	Te $5s^2 5p^4$	I $5s^2 5p^5$	Xe $5s^2 5p^6$
Cs $6s^1$	Ba $6s^2$	La $6s^2 5d^1$	Hf $6s^2 5d^2$	Ta $6s^2 5d^3$	W $6s^2 5d^4$	Re $6s^2 5d^5$	Os $6s^2 5d^6$	Ir $6s^2 5d^7$	Pt $6s^2 5d^8$	Au $6s^2 5d^9$	Hg $6s^2 5d^{10}$	Tl $6s^2 6p^1$	Pb $6s^2 6p^2$	Bi $6s^2 6p^3$	Po $6s^2 6p^4$	At $6s^2 6p^5$	Rd $6s^2 6p^6$
Fr $7s^1$	Ra $7s^2$	Ac $7s^2 6d^1$															

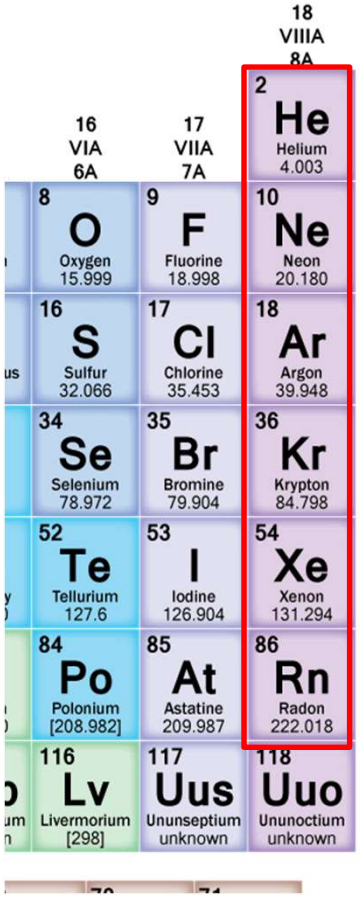
La $6s^2 5d^1$	Ce $6s^2 5d^1 4f^1$	Pr $6s^2 4f^3$	Nd $6s^2 4f^4$	Pm $6s^2 4f^5$	Sm $6s^2 4f^6$	Eu $6s^2 4f^7$	Gd $6s^2 5d^1 4f^7$	Tb $6s^2 4f^9$	Dy $6s^2 4f^{10}$	Ho $6s^2 4f^{11}$	Er $6s^2 4f^{12}$	Tm $6s^2 4f^{13}$	Yb $6s^2 4f^{14}$	Lu $6s^2 5d^1 4f^{14}$
Ac $7s^2 6d^1$	Th $7s^2 6d^2$	Pa $7s^2 6d^1 5f^1$	U $7s^2 6d^1 5f^3$	Np $7s^2 6d^1 5f^1$	Pu $7s^2 5f^6$									

Figure 1.3 The periodic table of naturally occurring elements showing the electronic configuration of the elements. Only the last orbitals filled are shown, thus each element has electrons in the orbitals of all previous group 18 elements (noble gases) in addition to those shown. Superscripts indicate the number of electrons in each subshell.

Geochemistry, First Edition. William M. White.
© 2013 John Wiley & Sons, Ltd. Published 2013 by John Wiley & Sons, Ltd.

ELECTRON CONFIGURATION CONTROLS CHEMISTRY

- Only electrons in the outermost orbitals participate in chemical bonding.
- Elements that share the same configuration of electrons in their outer shells have similar chemical properties.
- Groups in the periodic table have the same outer shell electron configuration.
- The general rule is that filled outer p shells are the most stable.



$1s^2$

$2s^2\ 2p^6$

$3s^2\ 3p^6$

$4s^2\ 4p^6$

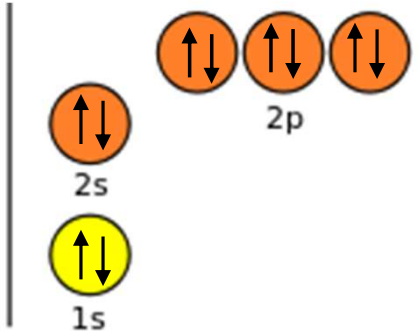
$5s^2\ 5p^6$

$6s^2\ 6p^6$

Noble gases:

- all have outer shells with s^2p^6 electron configuration
- they all have similar chemical behaviour
- filled outer p shells mean they are unreactive and don't form chemical bonds

Ne (Z=10) – ten electrons



ELECTRON CONFIGURATION CONTROLS CHEMISTRY



$2s^2\ 2p^5$

$3s^2\ 3p^5$

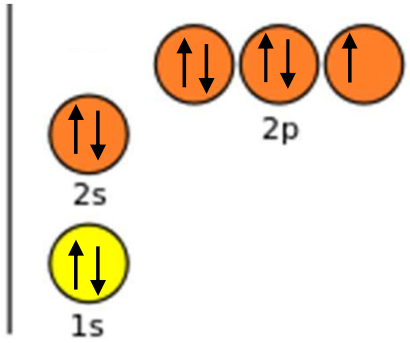
$4s^2\ 4p^5$

$5s^2\ 5p^5$

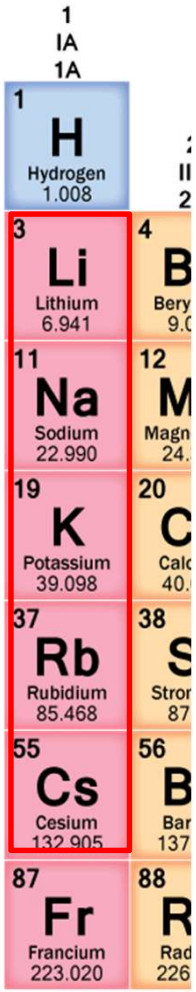
Halogen elements:

- all have outer shells with s^2p^5 electron configuration
- they all have similar chemical behaviour
- like to add an electron to make a filled outer shell (like a noble gas)
- then the atom has one more electron than the number of protons
- result is an ion (atom) with a negative charge (e.g. F^-)

F (Z=9)



ELECTRON CONFIGURATION CONTROLS CHEMISTRY



$2s^1$

$3s^1$

$4s^1$

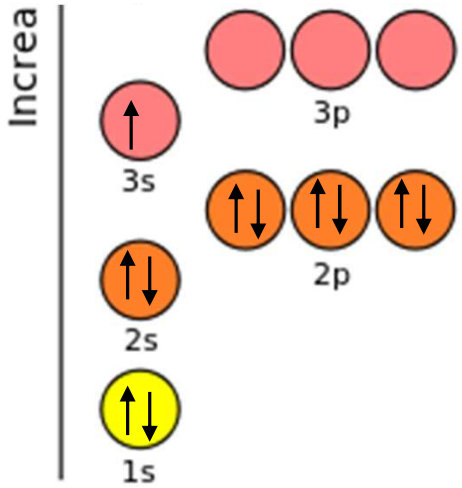
$5s^1$

$6s^1$

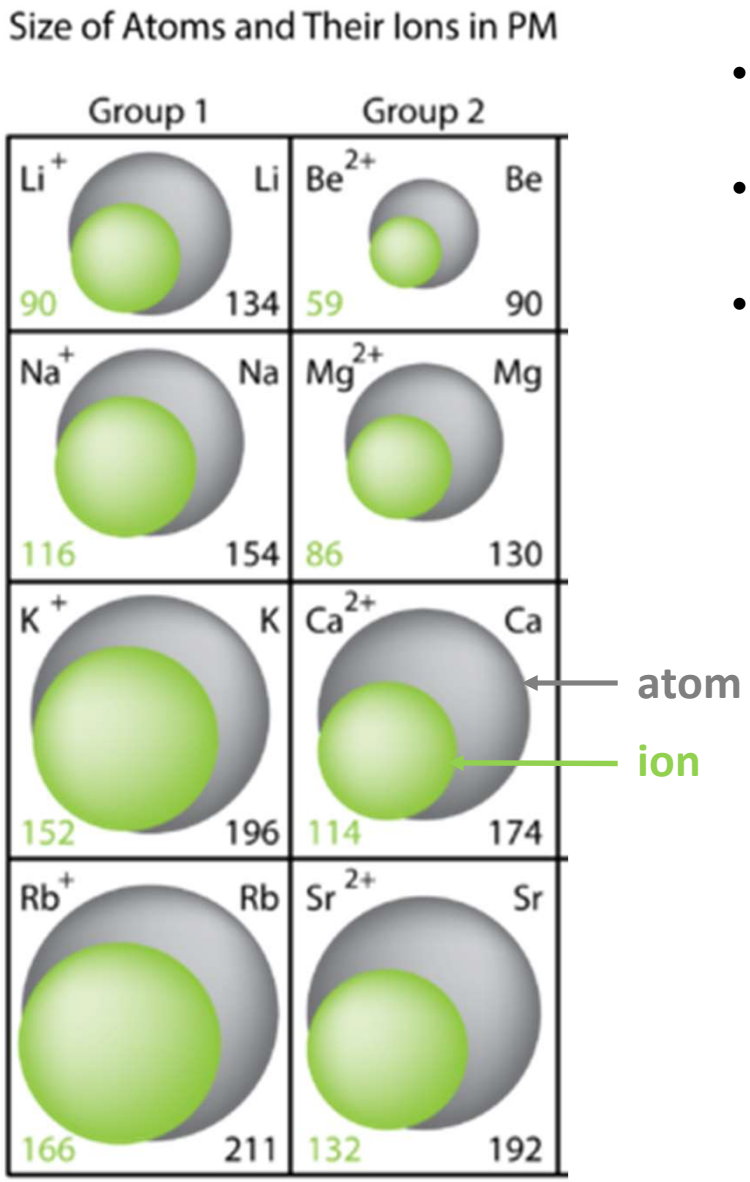
Alkali elements (Group 1):

- all have outer shells with s^1 electron configuration
- they all have similar chemical behaviour
- the s electron is easily lost to make a filled outer shell (like a noble gas)
- then the atom has one less electrons than the number of protons
- result is an ion (atom) with a positive charge (e.g. Na^+)

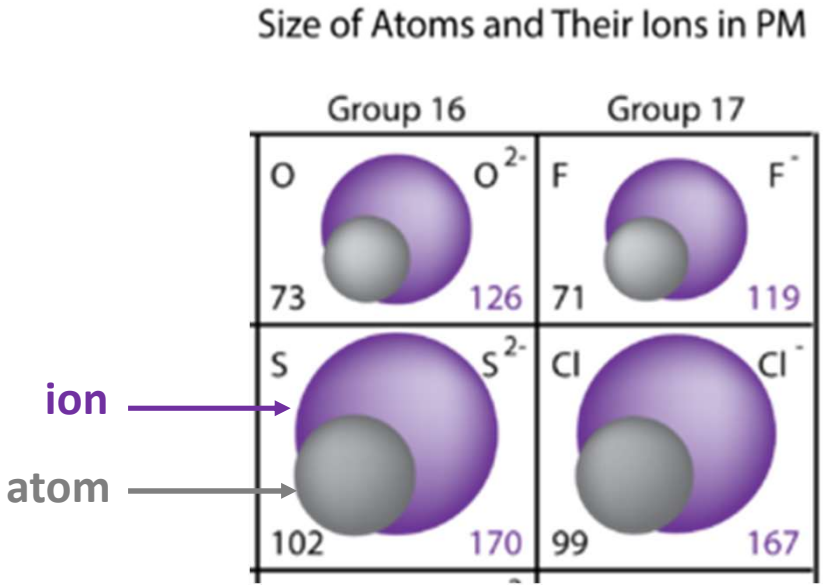
Na (Z=11)



Changes going down a group: atomic and ionic radii



- Radii of neutral atoms and ions increase as we go down a group
- Important when we consider crystal structures
- Ions with positive charge are smaller than their neutral atoms

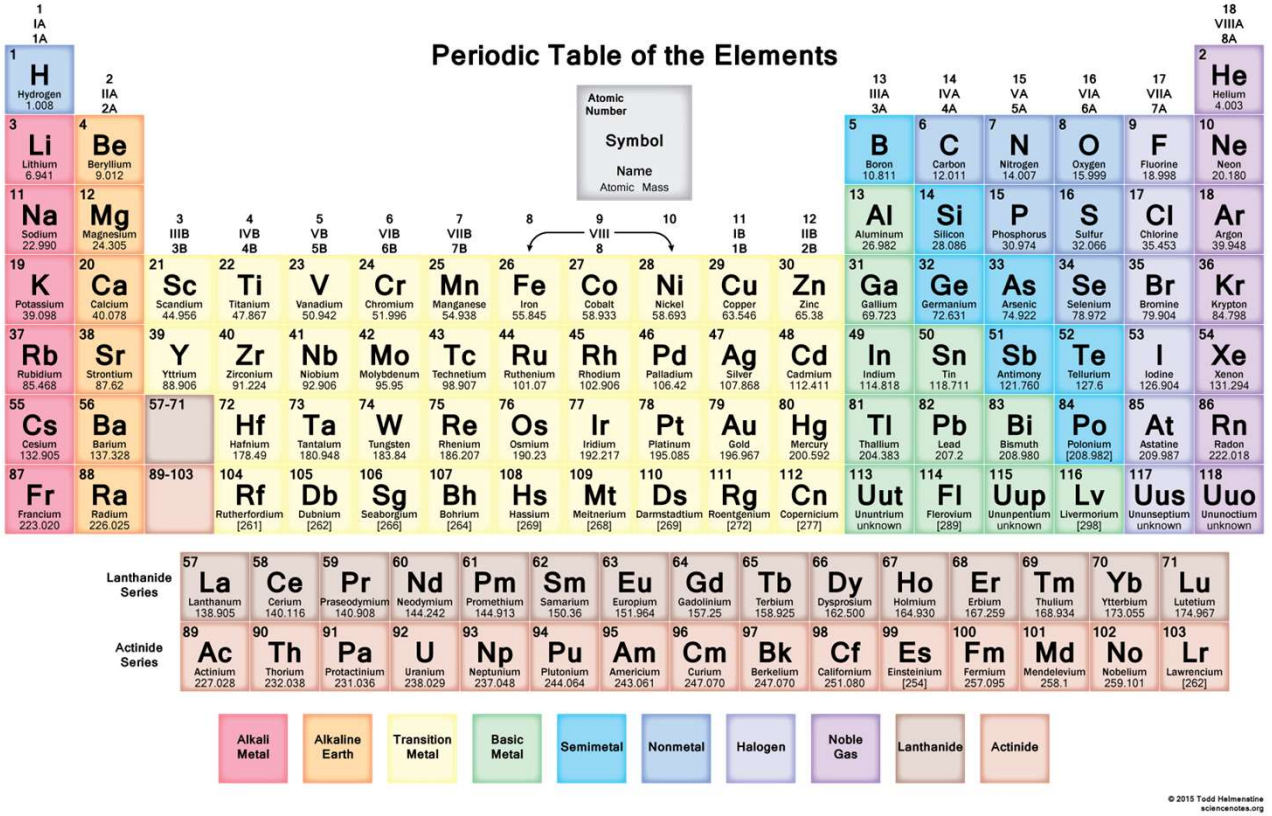


- Ions with negative charge are much larger than their neutral atoms

https://courses.lumenlearning.com/cheminter/chapter/ionic-radii/

Chemistry of the Earth, Lecture 2:

The Periodic Table – A Geochemist's View



Part 2: Chemical bonding in minerals

IMPORTANCE OF ELECTRON CONFIGURATION IN THE EARTH

Electron configurations control the behaviour of elements in solids (topic for today):

- Mineral lattice structure
- Bonding type (which controls mineral properties such as hardness)
- Mineral compositions
- Trace element substitutions

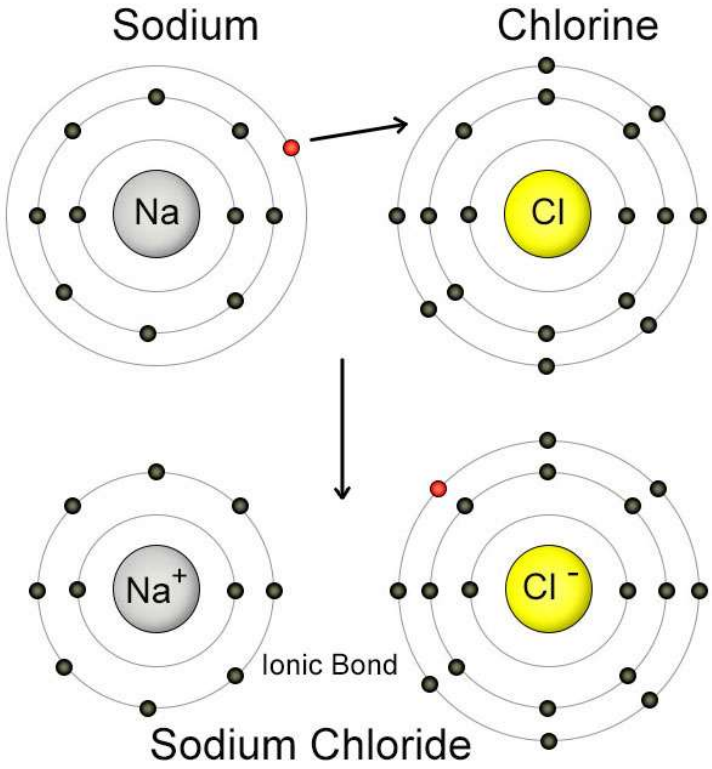
Electron configurations also control the behaviour of elements:

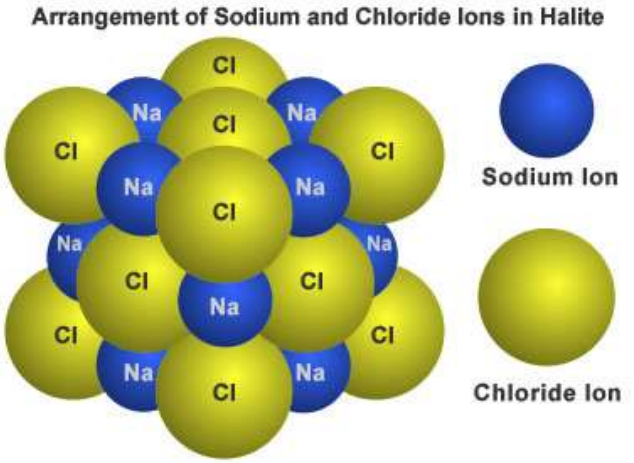
- In melts (magmas, lavas, the outer core)
- In fluids (hot fluids circulating in the mantle and crust)
- In waters (rivers, lakes, oceans)
- In gases (the atmosphere)
- In living organisms

CHEMICAL BONDING

Ionic bonding - elements form ions by losing or gaining electrons.

Example: NaCl (salt - mineral is halite)







Na: Z = 11 Cl: Z = 17

$1s^2\ 2s^2\ 2p^6\ 3s^1$ $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$

Wikipedia

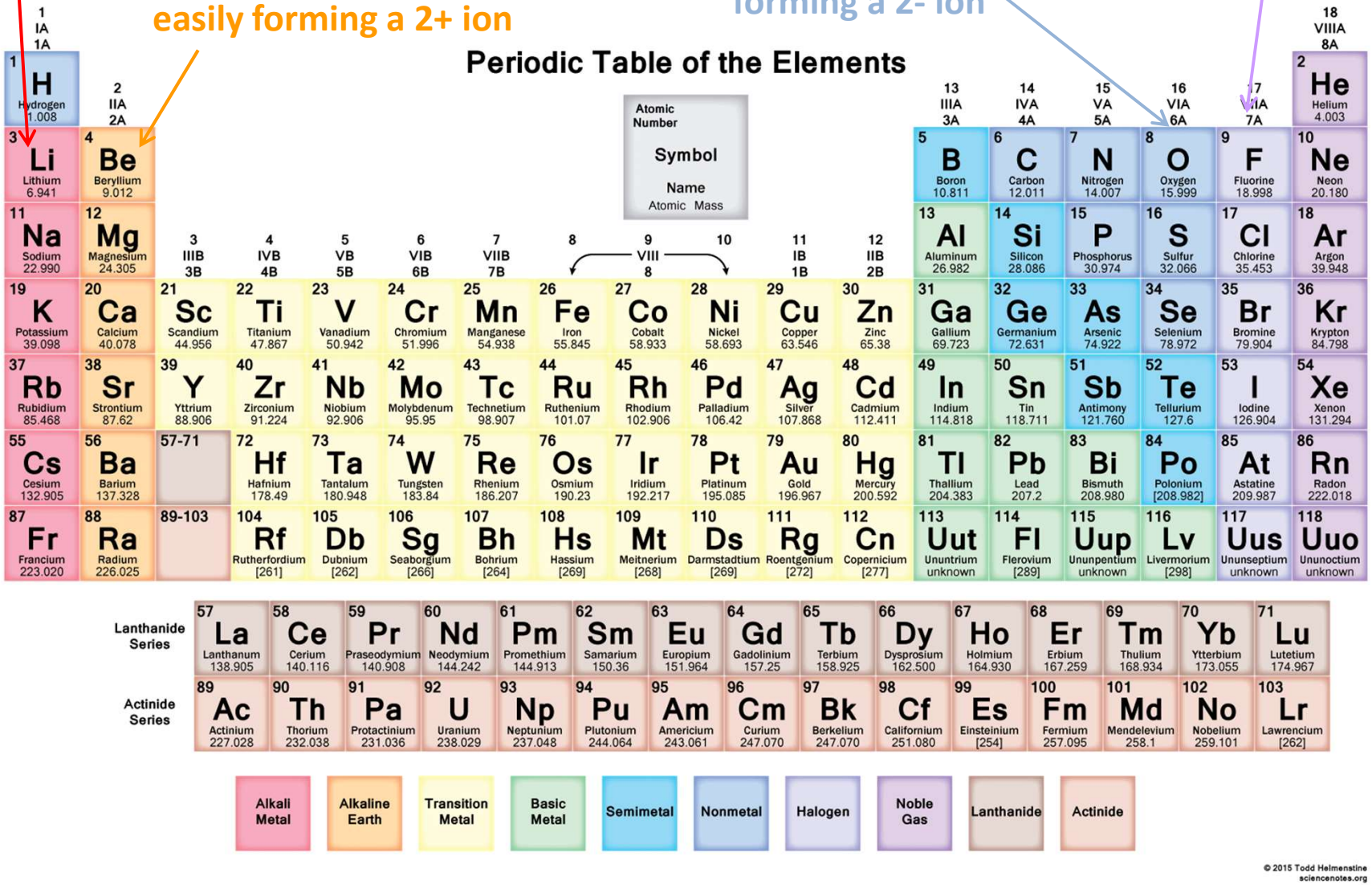
http://gardenandplate.com/sodiumchloride.html

s^1 – 1 electron is lost easily forming a 1+ ion

s^2 – 2 electrons are lost easily forming a 2+ ion

p^4 – 2 electrons can be added forming a 2- ion

p^5 – 1 electron is easily added forming a 1- ion



Ionic bonds also in Transition Metals, Lanthanides and Actinides

Terminology related to ions

CATIONS: Ions with a **POSITIVE** charge

ANIONS: Ions with a **NEGATIVE** charge

VALENCE: The number of electrons that an element will give up or accept.

Valence electrons: Electrons available for chemical bonding.

The concept of valence applies to ions with both positive and negative charges:

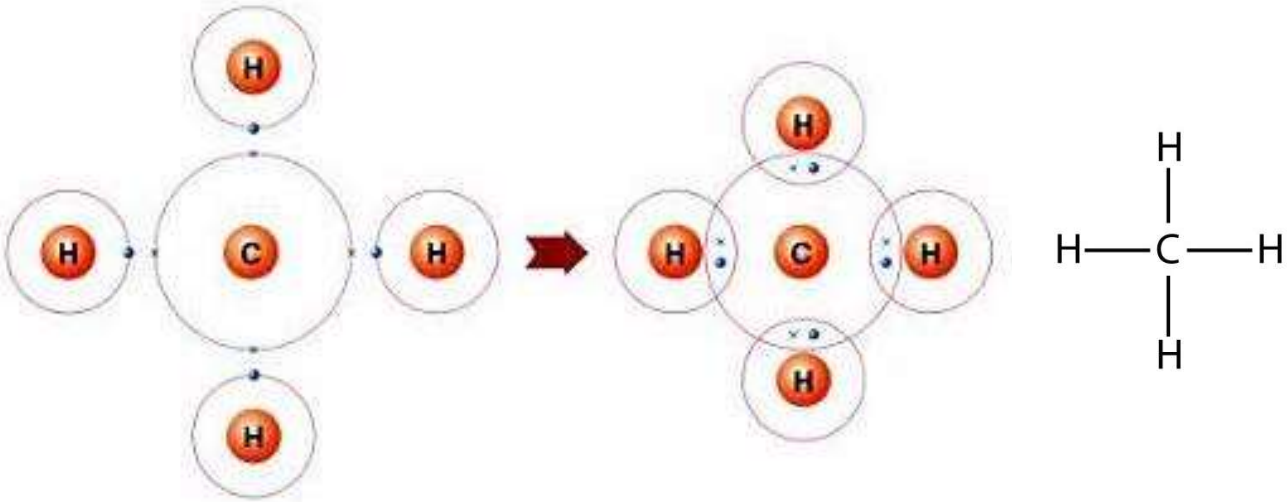
Monovalent ions:	Na^{+}	Cl^{-}	
Divalent ions:	Mg^{2+}	Fe^{2+}	O^{2-}
Trivalent ions:	Al^{3+}	Fe^{3+}	
Tetravalent ions:	Ti^{4+}		

CHEMICAL BONDING

Covalent bonding - atoms share electrons. No net change in charge.

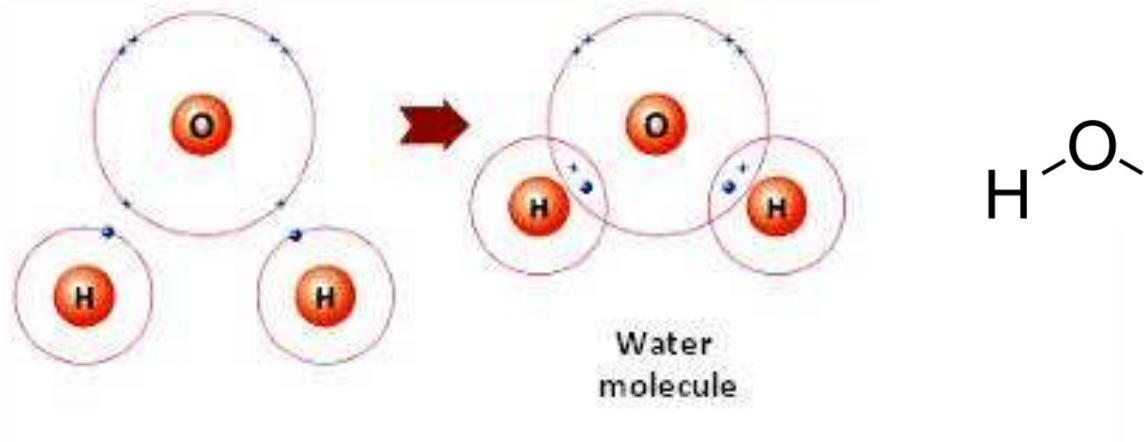
Methane: CH_4

Carbon has 4 electrons in 2nd shell.
Hard to lose 4, hard to gain 4.
Forms 4 covalent bonds.



Water: H_2O

Oxygen has 6 electrons in 2nd shell.
It can form O^{2-} ions or two covalent bonds.

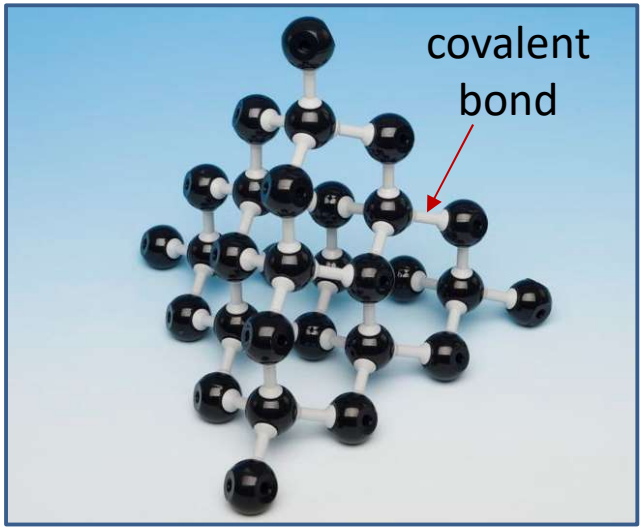


http://chemistry.tutorvista.com/physical-chemistry/covalent-bonding.html

CHEMICAL BONDING

Covalent bonding in minerals: strong bonds

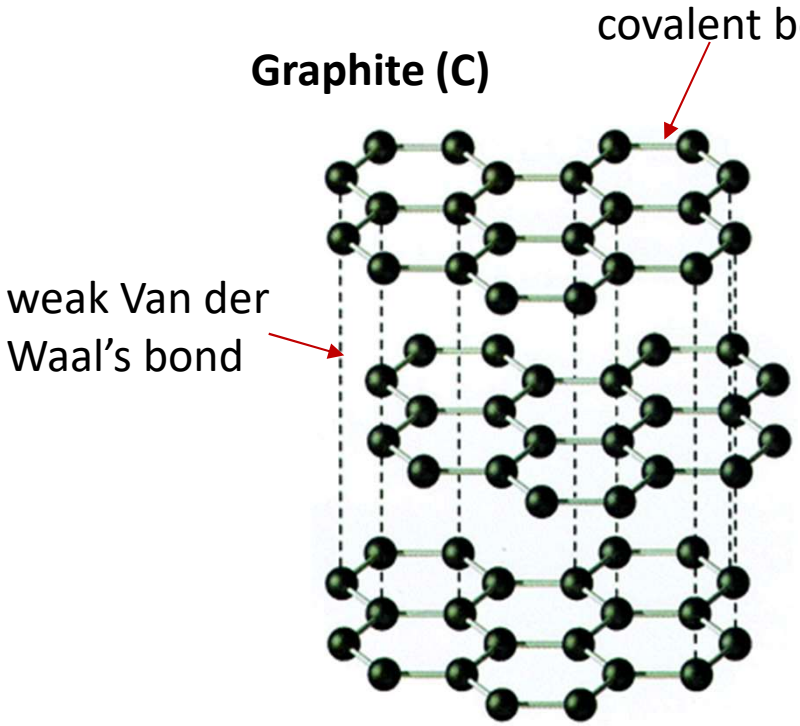
Diamond (C)





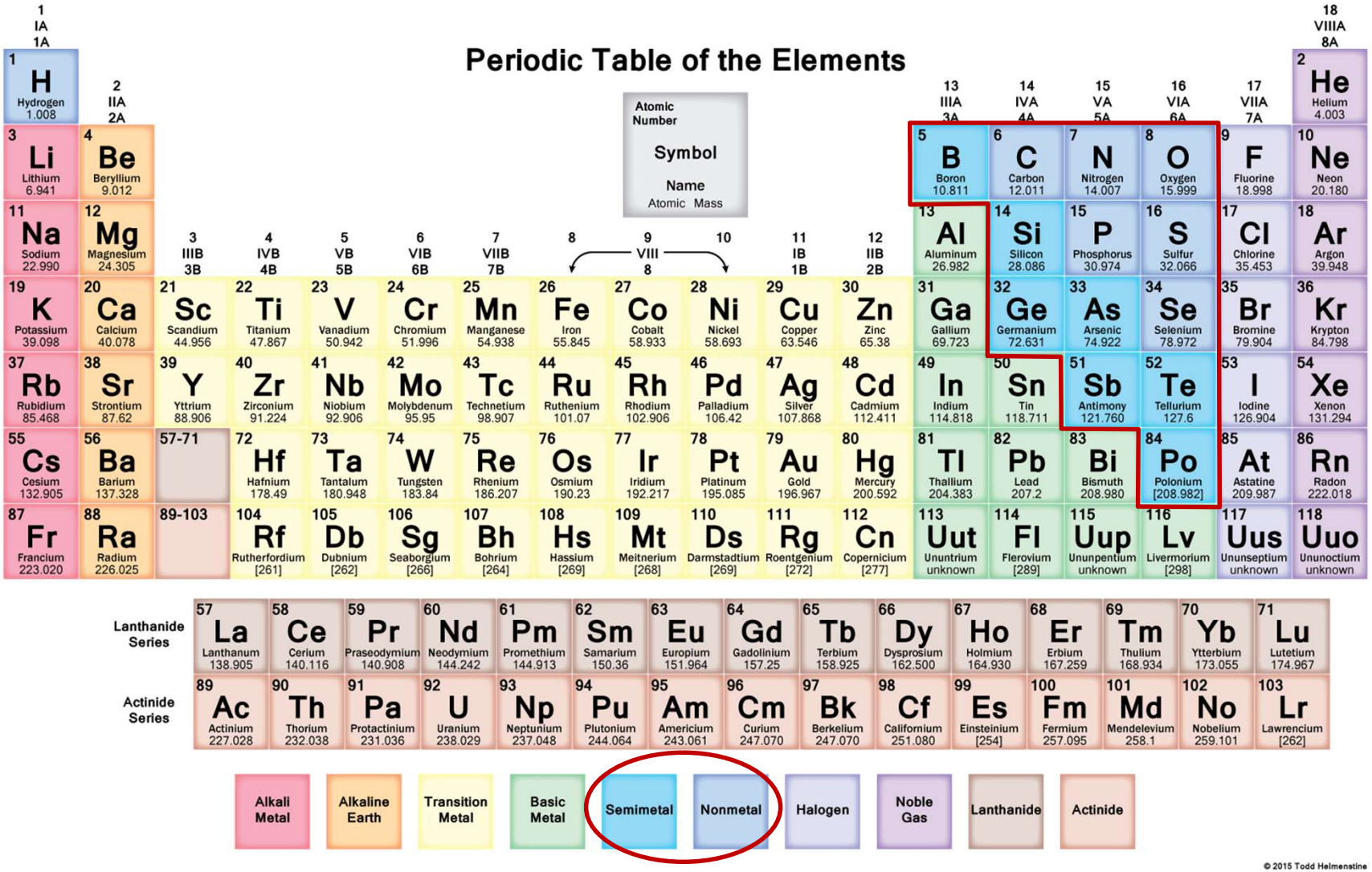
"Diamond - South Africa - Finsch Mine" by StrangerThanKindness

Graphite (C)



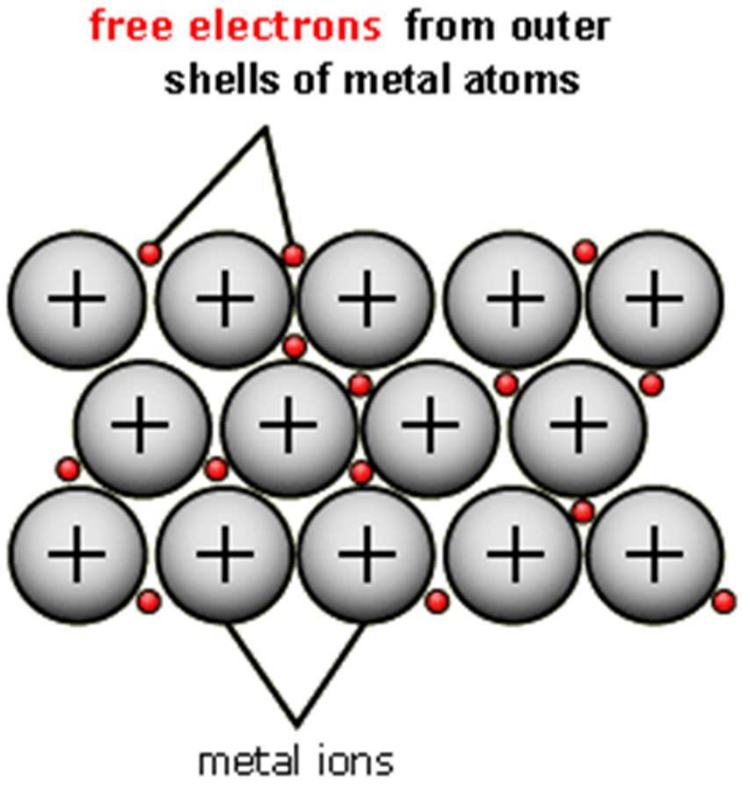


Semimetals and non-metals – covalent bonds



CHEMICAL BONDING

Metallic bonding – loosely bound cloud of shared electrons.
Common for elements with big outer shells and unfilled d orbitals.



"Native" metals

Copper (Cu)



Wikipedia

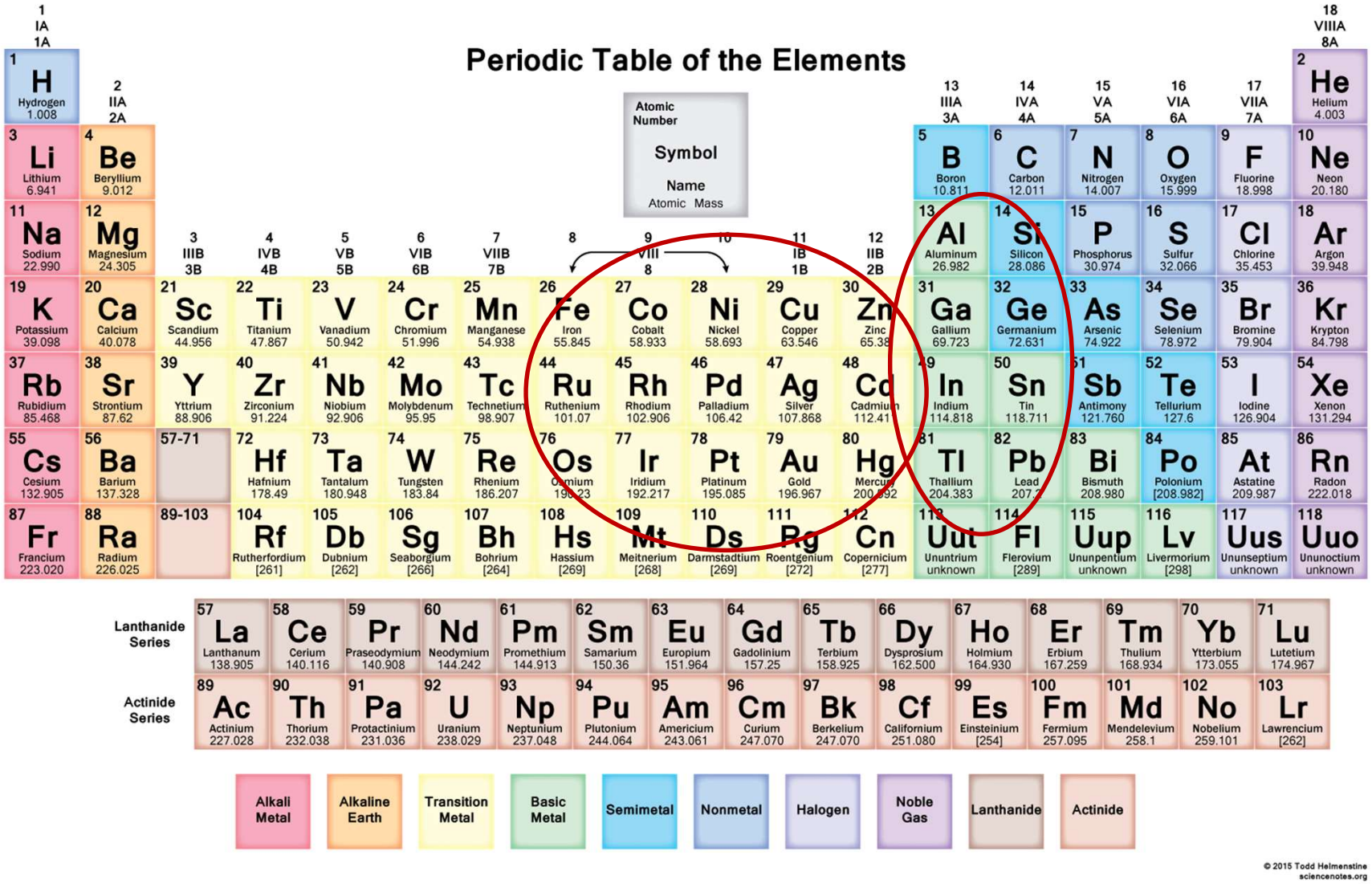
Gold (Au)



Staffordcountymuseum.com

http://www.bbc.co.uk/schools/gcsebitesize/science/add_ocr_gateway/periodic_table/metalsrev2.shtml

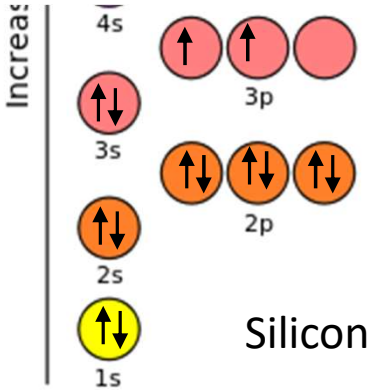
Metals – metallic bonding



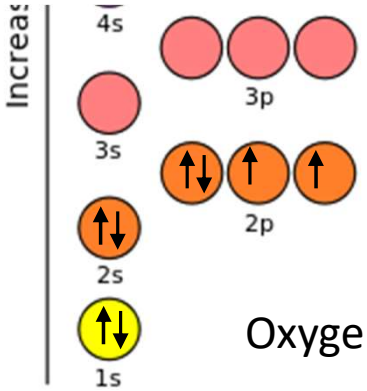
Many of the Transition Metals can form ionic bonds as well as metallic bonds

COMPLEX IONS: Silicate, carbonate, sulphate, phosphate

Silicate ion – SiO_4 tetrahedron – the building block of silicate minerals



Silicon (Z=14)



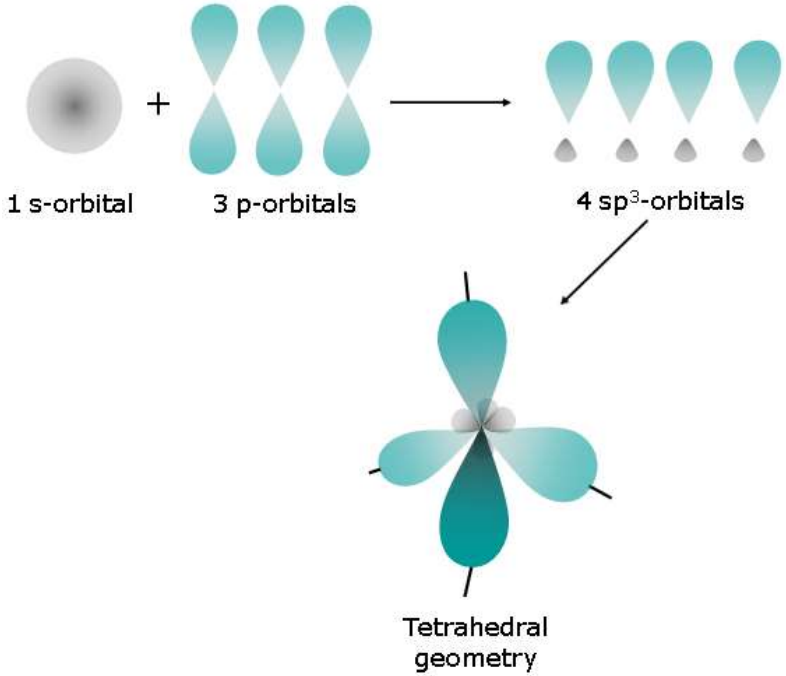
Oxygen (Z=8)

O: need to add two electrons to complete the second shell.

Si: need 4 electrons to complete the 3rd shell.

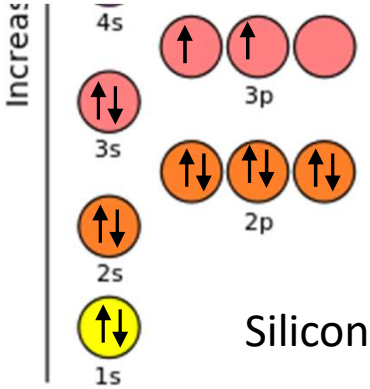
3s and 3p orbitals combine to make four equivalent "sp^3" orbitals, each of which forms a covalent bond with an oxygen atom.

The sp^3 orbitals point towards the corners of a tetrahedron.

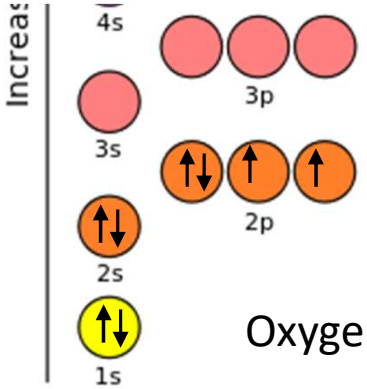


https://jahschem.wikispaces.com/electronic+structure

Silicate ion – SiO_4 tetrahedron – the building block of silicate minerals



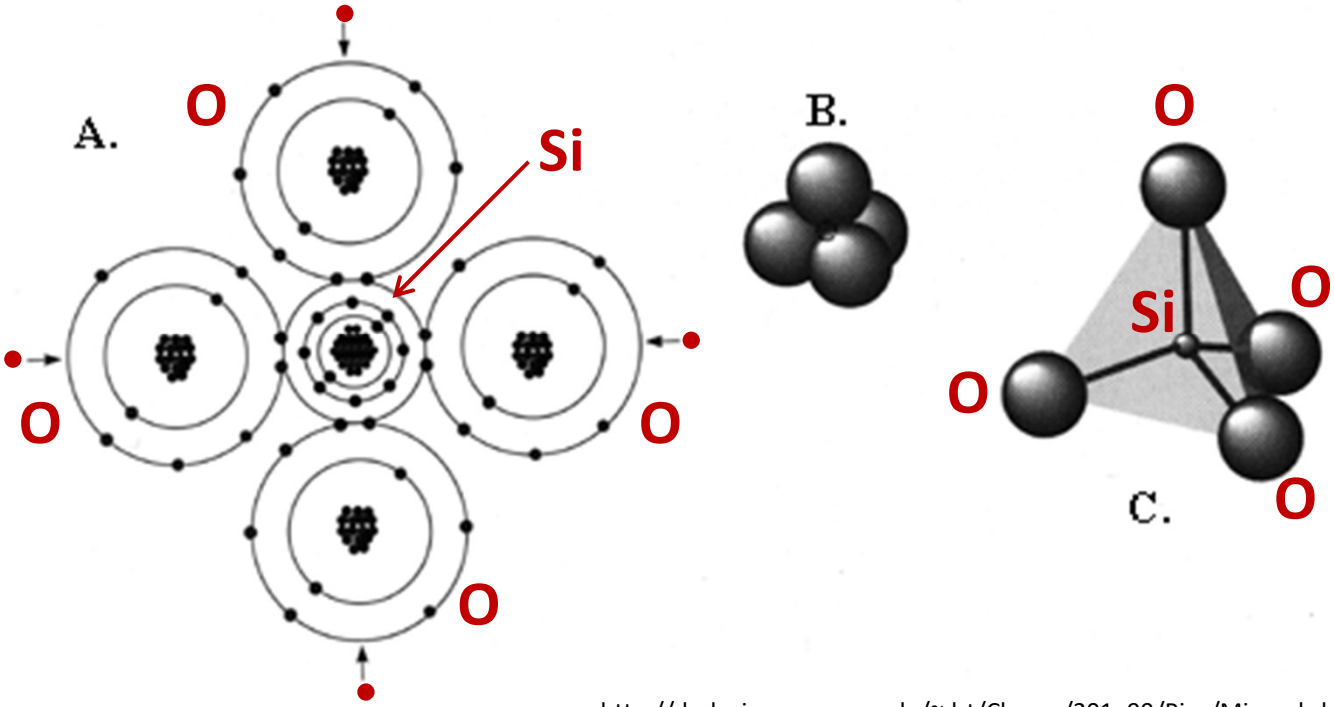
Silicon (Z=14)



Oxygen (Z=8)

Each oxygen atom also adds one extra electron to complete its outer shell.

Overall the unit is **$[SiO_4]^{4-}$**



http://darkwing.uoregon.edu/~drt/Classes/201_99/Rice/Minerals.html

Silicate ion – SiO_4 tetrahedron – the building block of silicate minerals

To make a mineral, need to balance the charge on $[SiO_4]^{4-}$ with cations.

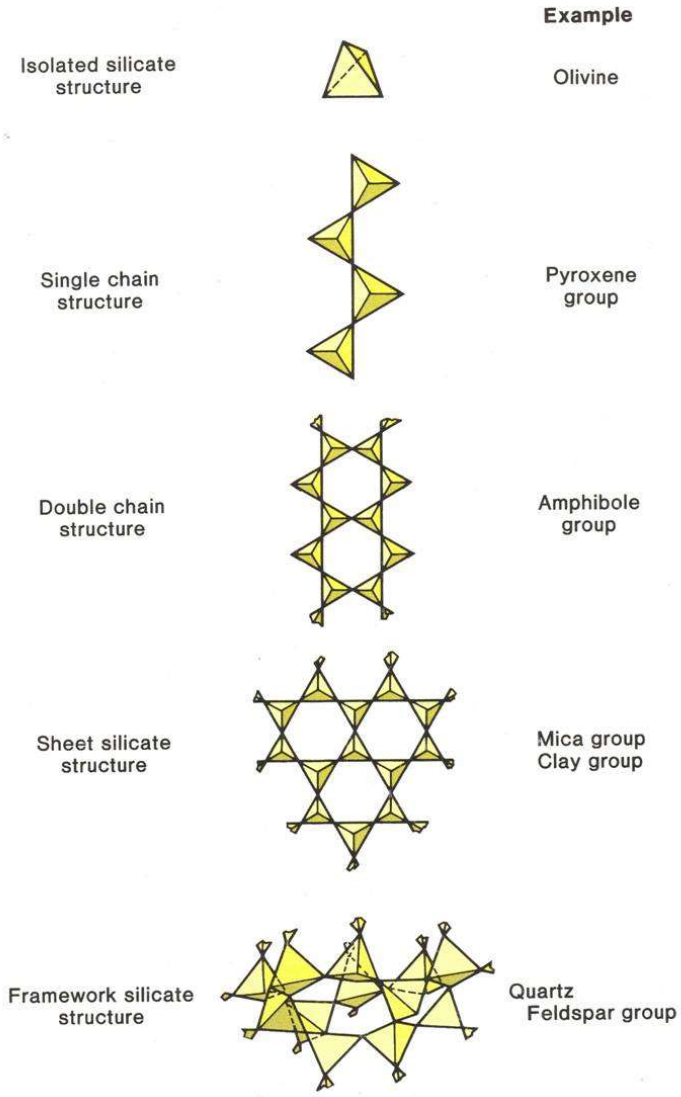
Example: Olivine, Mg_2SiO_4

Mg^{2+} Mg^{2+} $[SiO_4]^{4-}$



Mantle nodule - olivine is the green mineral

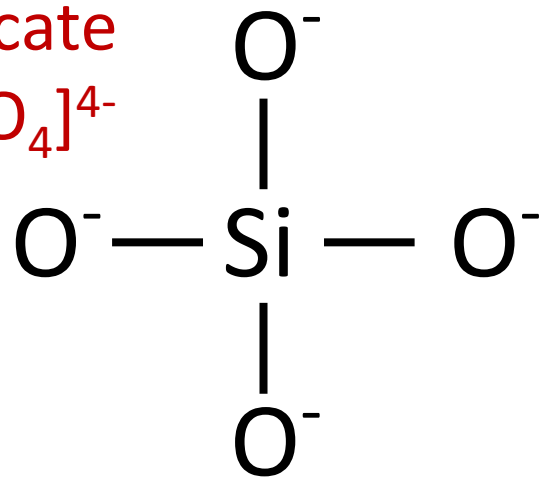
Silicate tetrahedra can be linked in many ways – examine this in Mineralogy



COMPLEX IONS: Silicate, carbonate, sulphate, phosphate

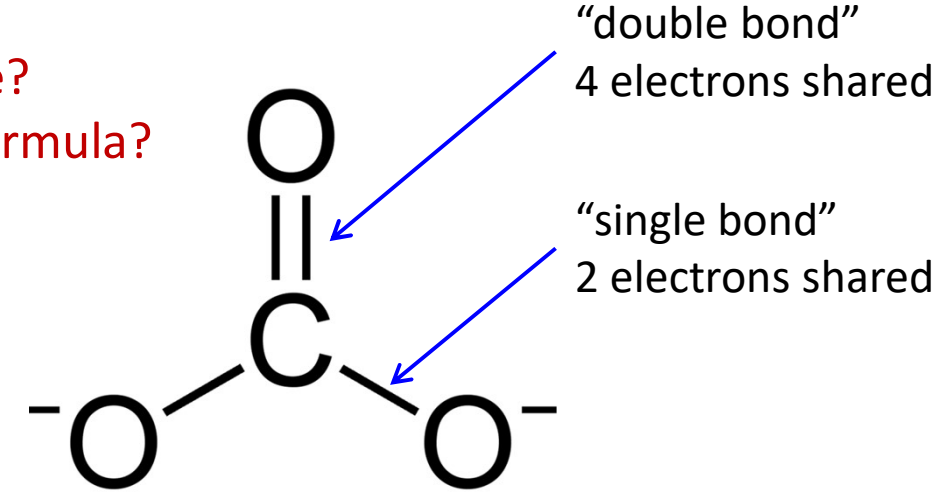
Silicate

$[SiO_4]^{4-}$



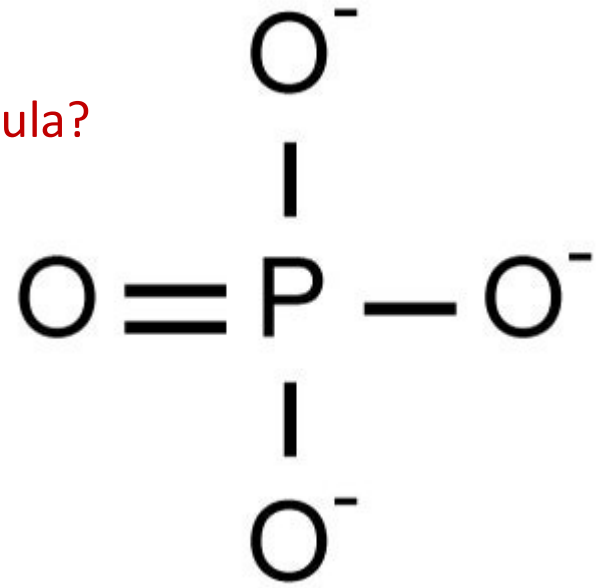
Name?

Ion formula?



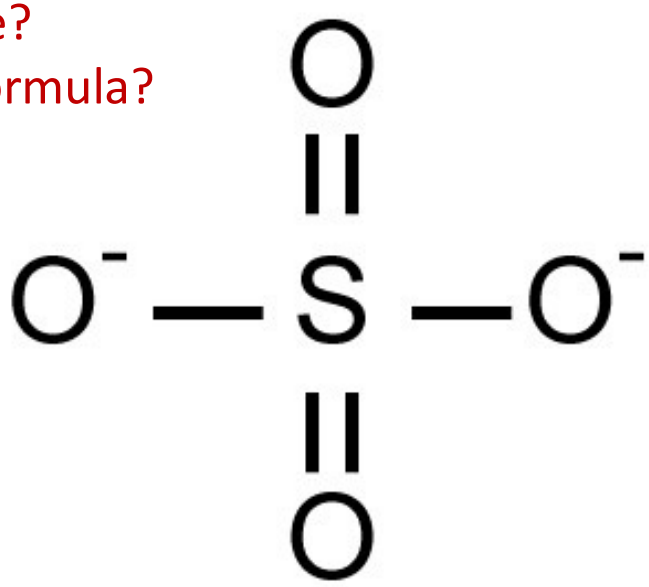
Name?

Ion formula?



Name?

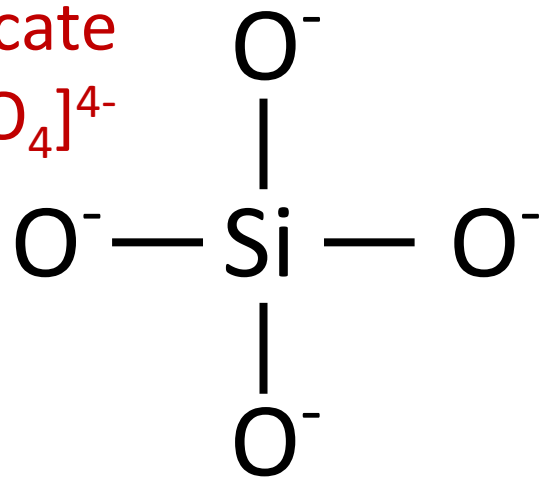
Ion formula?



COMPLEX IONS: Silicate, carbonate, sulphate, phosphate

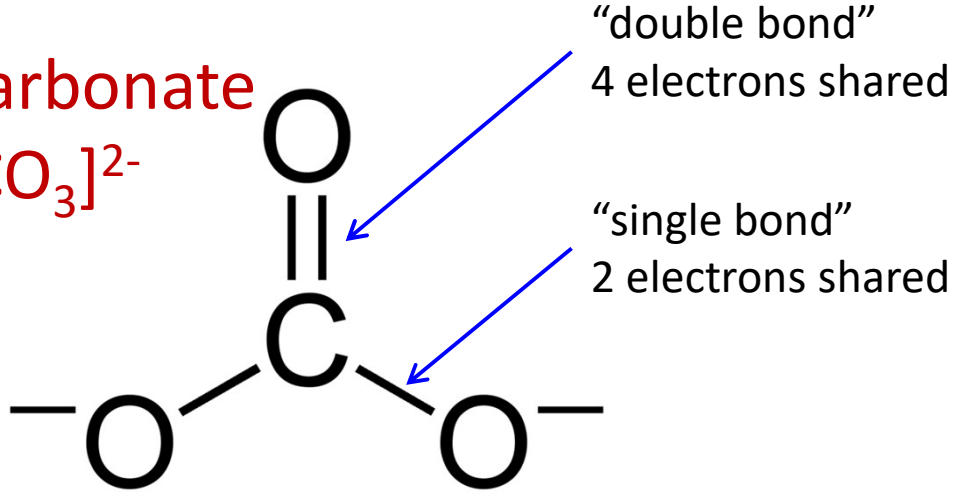
Silicate

$[SiO_4]^{4-}$



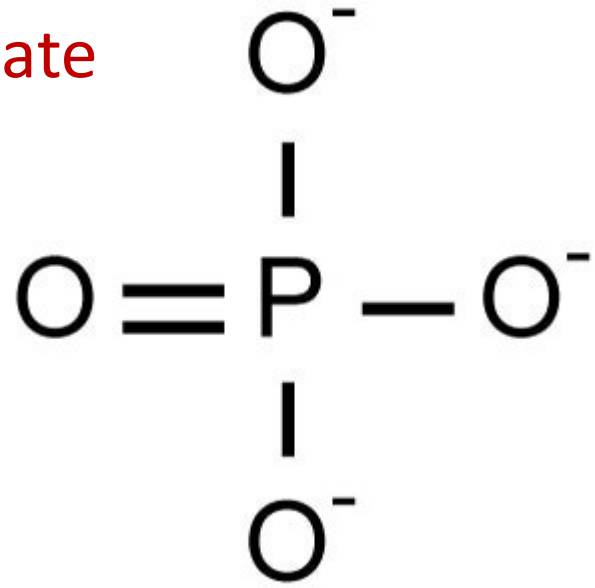
Carbonate

$[CO_3]^{2-}$



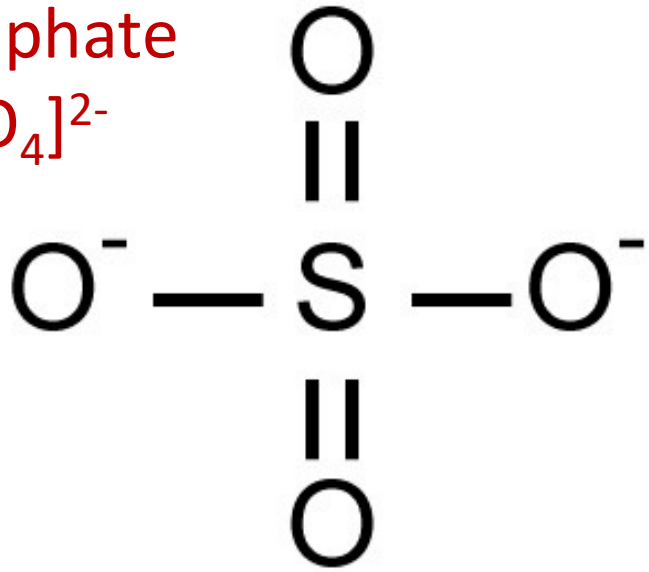
Phosphate

$[PO_4]^{3-}$



Sulphate

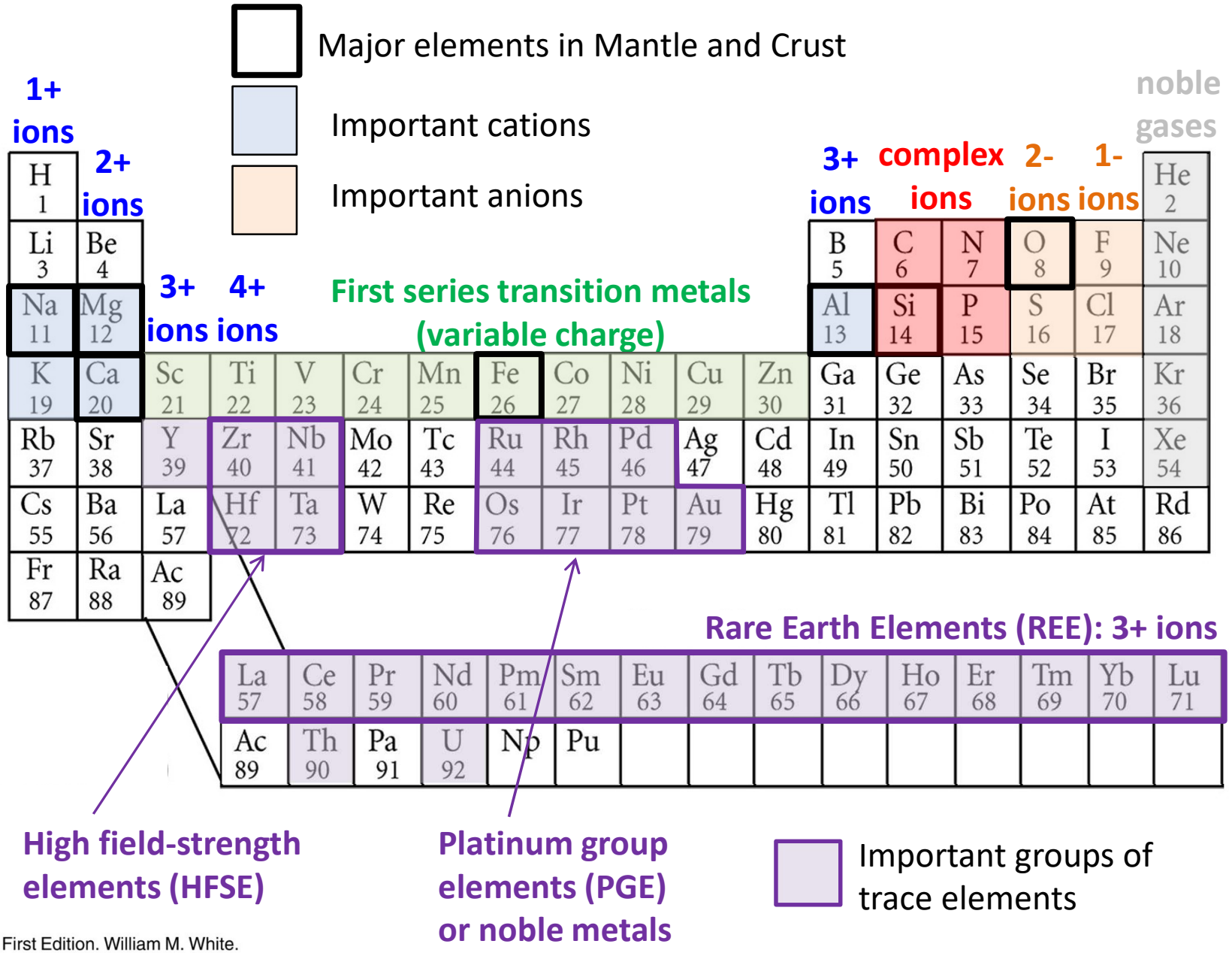
$[SO_4]^{2-}$



Practical 2:
Geochemist's Periodic Table

- Familiarize yourself with the Periodic Table, its groups and periods
- Review of atomic number, atomic mass, and isotopes
- Understand electron configurations and how they relate to the structure of the Periodic Table
- Practice writing mineral formulae based on charge balance

A GEOCHEMIST'S VIEW OF THE PERIODIC TABLE



Geochemistry, First Edition. William M. White.
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